

Nonlinear Wave Equations

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1 Overview

The following notes are a (brief) introduction into the theory of nonlinear wave equations, and mathematical fluids that I've written in preparation for the mathematics PhD oral examination at Stony Brook. These notes cover a few fundamental results on the (local) well-posedness of (non)linear wave equations, energy estimates, Strichartz estimates, and related results in harmonic analysis and fluids. The content reflects my most current understanding of the key ideas, and were written alongside discussions with my advisor, Professor John Anderson. Moreover, these notes are strongly based on the corresponding notes of Professors Holzegel, Lawrie, and Luk. It goes without saying that any errors in the following notes are entirely my responsibility.

Part I

Nonlinear Wave Equations

2 General Analytic Results

In this subsection, we will be introducing some general analytic results in functional analysis and harmonic analysis, that we will later use in our study of the (non)linear wave equations (and hyperbolic PDE, in general) [7].



2.1 Introductory Harmonic Analysis

In this subsection, we introduce key notions of harmonic analysis. In particular, we introduce the role of L^p spaces in harmonic analysis, then turn to *weak-type* estimates which will prove handy when we talk about estimates for (non)linear wave equations.

2.1.1 Motivation

We start with the following questions, that lead to our study of harmonic analysis. If $f \in L^2([0, 2\pi])$ is square integrable, then the L^2 Parseval relation tells us that

$$\sum_{n=-\infty}^{\infty} |a_n|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta, \quad (1)$$

where for each n ,

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta$$

are the **Fourier coefficients** of f such that

$$f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}.$$

Likewise, if $\{a_n\} \in \ell^2(\mathbb{Z})$, then there exists a unique $f \in L^2([0, 2\pi])$ such that $f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$ and (1) holds. If we assume $f \in L^p([0, 2\pi])$ or $\{a_n\} \in \ell^p(\mathbb{Z})$ for some $p \neq 2$, does the L^2 Parseval relation generalize to these cases? In general, one has the *Hausdorff-Young inequalities*,

$$(\sum |a_n|^q)^{1/q} \leq \left(\frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^p d\theta \right)^{1/p},$$

and

$$\left(\frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^q d\theta \right)^{1/q} \leq \left(\sum |a_n|^p \right)^{1/p}.$$

Thus in this subsection, we are interested in answering questions of this form, and setting up the framework for other types of estimates. In particular, we are interested in L^p -type estimates involving a function and its Fourier transform.

2.1.2 Riesz Interpolation Theorem

In this subsubsection, we encounter the **Riesz interpolation theorem**. When establishing various estimates for PDEs, it can be easier to establish corresponding inequalities for certain “endpoints”, and to “interpolate” in the corresponding estimates for all intermediate values. I.e., suppose one has the estimates $A_0 \leq B_0$ and $A_1 \leq B_1$. A natural question to ask is if for any $\theta \in (0, 1)$, $A_\theta \lesssim_\theta B_\theta$. In this subsubsection, we will focus on estimates for linear operators between L^p spaces, and we will give specific criteria under which interpolation in the above sense holds. In particular, we will answer the following question.

Suppose (p_0, q_0) and (p_1, q_1) are two pairs of indices with $1 \leq p_i, q_i \leq \infty$. Assume also that T is a linear operator and that,

$$\|Tf\|_{L^{q_0}} \leq M_0 \|f\|_{L^{p_0}} \quad \|Tf\|_{L^{q_1}} \leq M_1 \|f\|_{L^{p_1}}.$$

Does it follow necessarily that $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$ for other pairs (p, q) ?

We begin with the following technical lemma.

Lemma 2.1 (Three-Lines Lemma). *Suppose $\Phi(z)$ is a holomorphic function in the strip $S = \{z \in \mathbb{C} : 0 < \operatorname{Re}(z) < 1\}$, that is also continuous and bounded on the closure of S . If*

$$M_0 = \sup_{y \in \mathbb{R}} |\Phi(iy)| \quad \text{and} \quad M_1 = \sup_{y \in \mathbb{R}} |\Phi(1 + iy)|,$$

then

$$\sup_{y \in \mathbb{R}} |\Phi(t + iy)| \leq M_0^{1-t} M_1^t, \quad \text{for all } 0 \leq t \leq 1.$$

Proof. Let Φ be a holomorphic function in the strip $S = \{z \in \mathbb{C} : 0 < \operatorname{Re}(z) < 1\}$ that is also continuous and bounded on $\bar{S}$. If Φ is constant, then the claim follows trivially. So assume Φ is nonconstant. First, we assume that $M_0, M_1 = 1$. We also assume that $\Phi(x + iy) \rightarrow 0$ as $|y| \rightarrow \infty$. By the hypotheses on Φ , there exists some $M > 0$ such that

$$\sup_{z \in S} |\Phi(z)| = M.$$

In particular, it follows that there exists some sequence $\{z_n\}_1^\infty \subset S$ such that $\Phi(z_n) \rightarrow M$ as $n \rightarrow \infty$. But by the hypotheses, the z_n cannot converge to ∞ , and thus, must converge to some $z_0 \in S$. The maximum modulus principle and the fact that Φ is not constant forces $z_0 \in \partial S$. Thus by the hypotheses on M , one has that $M \leq 1$. Therefore, the proof concludes under these restrictive hypotheses.

Now assume that $M_0, M_1 = 1$, but that the decay rate is not necessarily satisfied. For each $\varepsilon > 0$, define

$$\Phi_\varepsilon(z) = \Phi(z) \exp(\varepsilon(z^2 - 1)).$$

Moreover, it follows that on the lines $\operatorname{Re}(z) = 0$ and $\operatorname{Re}(z) = 1$, $|\Phi_\varepsilon(z)| \leq 1$. Since Φ is bounded, it also follows that for each fixed $x \in [0, 1]$,

$$\sup_{x \in [0, 1]} |\Phi_\varepsilon(x + iy)| \xrightarrow{|y| \rightarrow \infty} 0.$$

Therefore, the first part of the proof shows that on $\bar{S}$, $|\Phi_\varepsilon| \leq 1$. Taking $\varepsilon \rightarrow 0$ concludes the proof.

For the final part of the proof, assume the least possible number of hypotheses on Φ , and define $\tilde{\Phi} = M_0^{z-1} M_1^{-z}$. Then $\tilde{\Phi}$ satisfies the hypotheses of the previous case so that $|\tilde{\Phi}| \leq 1$ in the strip, concluding the proof. $\square$

Remark 2.1. *The geometric intuition for this lemma is clear to see. We specify bounds for Φ on the two vertical lines $\operatorname{Re} \Phi = 0$ and $\operatorname{Re}(z) = 1$. Then under the right hypotheses, the above lemma gives bounds for any third vertical line $\operatorname{Re}(z) = t$ between $\operatorname{Re}(z) = 0$ and $\operatorname{Re}(z) = 1$.*

From this result, we now have the following interpolation theorem for L^p spaces.

Theorem 2.1 (Riesz-Thorin Interpolation Theorem). *Suppose T is a linear mapping from $L^{p_0} + L^{p_1}$ to $L^{q_0} + L^{q_1}$. Assume that T is bounded from L^{p_0} to L^{q_0} and from L^{p_1} to L^{q_1} :*

$$\begin{cases} \|Tf\|_{L^{q_0}} \leq M_0 \|f\|_{L^{p_0}}, \\ \|Tf\|_{L^{q_1}} \leq M_1 \|f\|_{L^{p_1}}. \end{cases}$$

Then T is bounded from L^p to L^q : $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$, whenever the pair (p, q) can be written as

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1} \quad \text{and} \quad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1}$$

for some t with $0 \leq t \leq 1$. Moreover, the bound M satisfies $M \leq M_0^{1-t} M_1^t$.

Proof. Recall that the space of simple functions is dense in L^p for all p . Thus, our strategy is as follows: (1) show that the result holds for all simple functions and for all $p < \infty$ and $q > 1$, (2) use the density of simple functions in L^p to show the result for all $f \in L^p$ ($1 \leq p < \infty$), (3) show the result holds for $p = 1$ and $q = \infty$, and (4) finally show the result holds for $p < \infty$ and $q = 1$. We will also use the fact that to show that $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$, it suffices to show that $|\int Tfg \, dx| \leq M \|f\|_{L^p} \|g\|_{L^{q'}}$ for all simple functions g with $\|g\|_{L^{q'}} = 1$ ¹.

- (1) First assume that $p < \infty$ and $q > 1$, and assume $f \in L^p$ is simple with $\|f\|_{L^p} = 1$. We wish to define some holomorphic function Φ that is bounded on $\operatorname{Re}(z) = 0, 1$ so that we can use the interpolation result for this function. Thus for $z \in \mathbb{C}$, define

$$\begin{aligned} f_z &= |f|^{\gamma(z)} \frac{f}{|f|}, & \gamma(z) &= p \left(\frac{1-z}{p_0} + \frac{z}{p_1} \right), \\ g_z &= |g|^{\delta(z)} \frac{g}{|g|}, & \delta(z) &= q' \left(\frac{1-z}{q'_0} + \frac{z}{q'_1} \right). \end{aligned}$$

These definitions are useful for the following reasons. First, it is straightforward to see that for any $t \in [0, 1]$, $f_t = f$ and $g_t = g$. Next, for $\operatorname{Re}(z) = 0$, we compute

$$\|f_z\|_{L^{p_0}}^{p_0} = \int |f|^{\gamma(z)p_0} \, d\nu = \int |f|^{p_0 \cdot \operatorname{Re}(\gamma(z))} = \int |f|^p = \|f\|_{L^p}^p = 1.$$

Likewise, for $\operatorname{Re}(z) = 1$, we compute

$$\|f_z\|_{L^{p_1}}^{p_1} = \int |f|^{\gamma(z)p_1} \, d\nu = \int |f|^{p_1 \cdot \operatorname{Re}(\gamma(z))} = \int |f|^p = \|f\|_{L^p}^p = 1.$$

Likewise, one has

$$\|g_z\|_{L^{q'_0}} = 1 \quad \text{and} \quad \|g_z\|_{L^{q'_1}} = 1.$$

Thus define

$$\Phi(z) = \int (Tf_z)g_z \, d\nu.$$

Since f, g are simple, assume

$$f_z = \sum |a_k|^{\gamma(z)} \frac{a_k}{|a_k|} \chi_{E_k} \quad \text{and} \quad g_z = \sum |b_j|^{\delta(z)} \frac{b_j}{|b_j|} \chi_{F_j},$$

¹Note: by $'$, we shall denote the conjugate exponent of an exponent.

where the E_k and F_j are sets of finite measure. This implies that

$$\Phi(z) = \sum_{j,k} |a_k|^{\gamma(z)} |b_j|^{\delta(z)} \frac{a_k}{|a_k|} \frac{b_j}{|b_j|} \left(\int T(\chi_{E_k}) \chi_{F_j} d\nu \right)$$

so that (a) Φ is holomorphic on the strip S , and bounded, continuous on $\bar{S}$. For $\operatorname{Re}(z) = 0$, one has

$$|\Phi(z)| \leq \int |Tf_z| |g_z| d\nu \leq \|Tf_z\|_{L^{q_0}} \|g\|_{L^{q'_0}} = \|Tf_z\|_{L^{q_0}} \leq M_0 \|f\|_{L^{p_0}} = M_0.$$

Likewise on $\operatorname{Re}(z) = 1$, one has

$$|\Phi(z)| \leq \int |Tf_z| |g_z| d\nu \leq \|Tf_z\|_{L^{q_1}} \|g\|_{L^{q'_1}} = \|Tf_z\|_{L^{q_1}} \leq M_1 \|f\|_{L^{p_1}} = M_1.$$

Therefore, by the three-lines lemma, one has that on the line $\operatorname{Re}(z) = t$ ($t \in [0, 1]$), $|\Phi(z)| \leq M_0^{1-t} M_1^t$. Thus this completes the proof.

- (2) Now fix some $f \in L^p$ for $1 \leq p < \infty$. Choose a sequence $\{f_n\}$ of simple functions in L^p so that $\|f_n - f\|_{L^p} \rightarrow 0$. Since $\|Tf_n\|_{L^q} \leq M \|f_n\|_{L^p}$, Tf_n is a Cauchy sequence in L^q . Thus it suffices to show that $Tf_n \rightarrow Tf$ a.e. To prove this, let us define,

$$f^U(x) := \begin{cases} f(x), & \{x : |f(x)| \geq 1\}, \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad f^L(x) := \begin{cases} f(x), & \{x : |f(x)| < 1\}, \\ 0, & \text{otherwise} \end{cases}$$

Then clearly $f = f^U + f^L$. Likewise, let $f_n = f_n^U + f_n^L$ for all n . Now wlog assume that $p_0 \leq p_1$ so that $p_0 \leq p \leq p_1$. Since $p_0 \leq p$, and $1 \leq p$,

$$\int |f^U|^{p_0} \leq \int |f^U|^p \leq \int |f|^p < \infty,$$

so that $f^U \in L^{p_0}$. Since $f_n \rightarrow f$ in L^p , it follows that $f_n^U \rightarrow f^U$ in L^{p_0} . Thus there exists a subsequence $\{Tf_{n_j}^U\}$ that converges to Tf^U pointwise a.e. Likewise, we have

$$\int |f^L|^{p_1} \leq \int |f^L|^p \leq \int |f|^p < \infty,$$

so that $f^L \in L^{p_1}$. Since $f_n \rightarrow f$ in L^p , it follows that $f_n^L \rightarrow f^L$ in L^{p_1} . Thus the sequence $f_{n_j}^L \rightarrow f^L$ in L^{p_1} which means that $\{Tf_{n_j}^L\}$ converges to Tf^L in L^{q_1} . This means there exists a subsubsequence $\{Tf_{n_{j_k}}^L\}$ that converges to Tf^L pointwise a.e.; by construction, $Tf_{n_{j_k}}^U$ converges to Tf^U pointwise a.e. Therefore, renaming this subsubsequence as $\{Tf_n\}$, we conclude that $Tf_n = Tf_n^U + Tf_n^L$ converges to Tf a.e. so that $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$.

- (3) Now consider the case $q = 1$ and $p = \infty$. Suppose first that $p = \infty$. Necessarily, it follows that $p_0 = p_1 = \infty$. Then Hölder's inequality concludes the proof.
- (4) Suppose $p < \infty$ and $q = 1$. Then $q_0 = q_1 = 1$. We can take $g_z = g$ for all z and argue as in the case for $q > 1$.

□

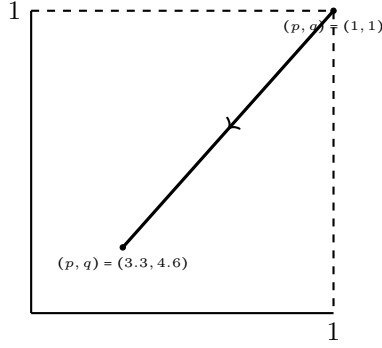
In particular, the following definition highlights the key essence of the Riesz-Thorin theorem.

Definition 2.1 (Operator of Type (p, q)). *Let $T : X \rightarrow Y$ be a linear operator that is initially defined on simple functions of X and maps such functions to functions on Y that are integrable on sets of finite measure. T is said to be an **operator of type** (p, q) if there exists $M > 0$ such that $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$ whenever f is simple.*

Since the space of simple functions is dense in every L^p space, it follows that if T is an operator of type (p, q) for some such pair, then the result extends easily to the entire L^p space for $p < \infty$ ($p = \infty$ if X has finite measure). We can denote the space of all such pairs (p, q) using a *Riesz diagram*.

Definition 2.2 (Riesz Diagram). *Let T be a linear operator as defined above. Then the **Riesz diagram** of T is the set of all points $(1/p, 1/q)$ in the unit square $[0, 1]^2$ such that T is of type (p, q) . We define $M_{x,y}$ to be the least M for which $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$ among all (p, q) such that T is of type (p, q) .*

For example, suppose T is a linear operator that satisfies the Riesz-Thorin Interpolation Theorem for $(p_0, q_0) = (1, 3.3)$, and $(q_0, q_1) = (1, 4.6)$. Then the Riesz diagram for T (as guaranteed by the Riesz-Thorin Interpolation Theorem) is the following,



We have the following corollary.

Corollary 2.1 (Riesz Convexity Theorem). *With T as before,*

- (a) *The Riesz diagram of T is a convex set.*
- (b) *$\log(M_{x,y})$ is a convex function on this set.*

Now we are interested in applications of this theorem.

Corollary 2.2 (Hausdorff-Young Inequality). *Let $X = [0, 2\pi]$ with the normalized Lebesgue measure $d\theta/(2\pi)$, and let $Y = \mathbb{Z}$ with its usual counting measure. Let T be the map defined by $Tf = \{a_n\}$, where a_n is the Fourier coefficient f :*

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta.$$

If $1 \leq p \leq 2$ and $1/p + 1/q = 1$, then

$$\|Tf\|_{L^q(\mathbb{Z})} \leq \|f\|_{L^p([0, 2\pi])}.$$

Proof. This is a classic example of a proof using interpolation. First assume $p_0, q_0 = 2$. Then the proof follows directly from Plancherel's theorem. Now assume $p_1 = 1, q_1 = \infty$. □

2.2 Distributions and Generalized Functions

In this subsection, we introduce the notion of *distributions*, which will become essential in our future study of nonlinear wave equations. Classically, we are used to the idea of a function f (on $\mathbb{R}^n$) as assigning a definite value $f(x)$ for every $x \in \mathbb{R}^n$. But we also saw in measure theory that it suffices to define a function up to a set of measure zero (i.e., we can allow f to remain undefined on points contained in a set of measure zero).² Now, we will see that it suffices to relax this definition even more in certain cases, leading to the idea of **distributions** or **generalized functions**. Rather than assigning a definite value at each point in its domain, we define a distribution by its averages taken with respect to a certain class of test functions (e.g., $C_c^\infty(\mathbb{R}^n)$). As we will see in Section §.3, one might not have classical solutions of desired regularity (in the sense of ordinary functions) for various partial differential equations. However, there could exist generalized functions, that although do not necessarily have the regularity to satisfy the original differential equations, that satisfy other properties that a normal classical solution would have. For example, consider the one-dimensional wave equation,

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0.$$

Multiplying both sides by an arbitrary smooth function, $\varphi(x, t)$ with compact support, and integrating one has,

$$\int_{\mathbb{R}} \int_{\mathbb{R}} \frac{\partial u}{\partial t} \varphi(x, t) dt dx + \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{\partial u}{\partial x} \varphi(x, t) dt dx.$$

Using Fubini's theorem, and integration by parts, one obtains

$$\int_{\mathbb{R}} \int_{\mathbb{R}} u(t, x) \frac{\partial \varphi}{\partial t} dt dx + \int_{\mathbb{R}} \int_{\mathbb{R}} u(t, x) \frac{\partial \varphi}{\partial x} dt dx = 0.$$

Thus any solution to the wave equation satisfies the above equation as long as it is at least C^1 . However, it is possible that there exist functions u that satisfy the above equation for all such φ but is possibly not differentiable (so that u does not satisfy the wave equation, itself). Hence, as we will see, distributions enable us to make sense of such u (which we shall call *weak solutions* to a differential equation).

Thus we have the following definitions.

Definition 2.3 (Test Functions on Ω). *Fix an open set Ω in $\mathbb{R}^d$. The **test functions** are those functions $\varphi \in C_c^\infty(\Omega)$ (or equivalently, $C_0^\infty(\Omega)$) - the set of all smooth functions with compact support in Ω . We denote the space of all test functions on Ω by the symbol $\mathcal{D}(\Omega)$.*

Definition 2.4 (Convergence in $\mathcal{D}(\Omega)$). *Let $\{\varphi_n\} \subset \mathcal{D}(\Omega)$ and $\varphi \in \mathcal{D}$. φ_n is said to converge to φ if the supports of φ_n are contained in a common compact set, and for every multiindex α , one has that $\partial_x^\alpha \varphi_n \rightarrow \partial_x^\alpha \varphi$ uniformly in x as $n \rightarrow \infty$.*

Definition 2.5 (Distributions on Ω). *A **distribution** F on Ω is a complex-valued linear functional $\varphi \mapsto F(\varphi)$, defined for all $\varphi \in \mathcal{D}$, that is continuous in the sense that $F(\varphi_n) \rightarrow F(\varphi)$ whenever $\varphi_n \rightarrow \varphi$ in $\mathcal{D}$. We shall denote the vector space of all distributions on Ω by $\mathcal{D}^*(\Omega)$.*

For example, all signed Radon measures on an open subset Ω , and all locally integrable function on Ω are examples of distributions on Ω . Our ultimate goal is to “calculus” using distributions. For smooth functions f , one of the most important results we have (as we shall see in the Section §.3) is **integration by parts**: for any $f \in C^\infty(\Omega)$ and $\varphi \in \mathcal{D}(\Omega)$, a simple inductive argument yields,

$$\int_{\Omega} (\partial_x^\alpha f) \varphi(x) dx = (-1)^{|\alpha|} \int_{\Omega} f (\partial_x^\alpha \varphi) dx,$$

where the boundary terms are identically zero because φ has compact support. Thus, generalizing this definition,

²which is to say that a function is really just an equivalence class of functions that agree pointwise a.e.

Definition 2.6 (Derivative of a Distribution). *If $F \in \mathcal{D}^*(\Omega)$, for any multiindex α , one defines the **distributional derivative of F** , $(\partial_x^\alpha F)$ as follows: for any $\varphi \in \mathcal{D}(\Omega)$,*

$$(\partial_x^\alpha F)(\varphi) = (-1)^{|\alpha|} F(\partial_x^\alpha \varphi). \quad (2)$$

Thus while an arbitrary locally integrable function f may not have sufficient regularity to talk about its classical derivatives, we can certainly talk about its partial derivatives as distributions (which exist).

We can generalize some basic properties of functions to distributions.

Definition 2.7 (Some Basic Operations on Distributions). *Let Ω be an open subset of $\mathbb{R}^d$, and let $F \in \mathcal{D}^*$. Then we define the following.*

1. *If $\psi \in C^\infty(\Omega)$, then the product $\psi \cdot F$ is defined by $(\psi \cdot F)(\varphi) := F(\psi\varphi)$ for every $\varphi \in \mathcal{D}$. It stands that $\psi \cdot F \in \mathcal{D}^*$.*
2. *Let $\tau_\cdot$ denote the translation operator: $\tau_h f(x) := f(x - h)$ for any $h \in \mathbb{R}^d$. Then the corresponding definition for distributions is,*

$$\tau_h(F)(\varphi) = F(\tau_{-h}\varphi), \quad \forall \varphi \in \mathcal{D}.$$

3. *If L is a nonsingular linear transformation on $\mathbb{R}^d$, then if $f_L(x) := f(Lx)$ for functions, on distributions, one has*

$$F_L(\varphi) = |\det L|^{-1} \cdot F(\varphi_{L^{-1}}), \quad \forall \varphi \in \mathcal{D}.$$

Another property we would like to generalize is convolution of a distribution with a test function. We make the following definitions:

Definition 2.8 (Convolution of a Distribution with Test Functions). *Let $F \in \mathcal{D}(\mathbb{R}^d)$, and $\psi \in C_c^\infty(\mathbb{R}^d)$. Then we make the definitions,*

1. *(As a function): $(F * \psi)(x) := F(\tilde{\psi}_x)$, where $\tilde{\psi}_x := \psi(x - y)$,*
2. *(As a distribution): $(F * \psi)(\varphi) := F(\tilde{\psi} * \varphi)$, for all $\varphi \in \mathcal{D}(\mathbb{R}^d)$, with $\tilde{\psi} \equiv \tilde{\psi}_0$.*

To see where these definitions could come from, assume first that F is a locally integrable function. Then one has that

$$(F * \psi)(x) := \int_{\mathbb{R}^d} F(x - y)\psi(y) dy = \int_{\mathbb{R}^d} F(y)\psi(x - y) dy.$$

This leads naturally to the first definition. On the other hand, for any $\varphi \in \mathcal{D}(\mathbb{R}^d)$, if F is a locally integrable function, then

$$\begin{aligned} (F * \psi)(\varphi) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)\psi(x - y)\varphi(x) dy dx \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)\psi(x - y)\varphi(x) dx dy \quad (\text{by Fubini's Theorem}) \\ &= \int_{\mathbb{R}^d} F(y)(\tilde{\psi} * \varphi)(y) dy \\ &= F(\tilde{\psi} * \varphi). \end{aligned}$$

This leads naturally to the second definition.

Proposition 2.1 (Equivalence of Definitions of Convolution). *Suppose F is a distribution and $\psi \in \mathcal{D}$. Then*

1. *The two definitions of $F * \psi$ given above coincide.*
2. *The distribution $F * \psi$ is a C^∞ function.*

We can also talk about distributions converging in $\mathcal{D}^*$.

Definition 2.9 (Weak Convergence of Distributions). *A sequence of distributions $\{F_n\} \subset \mathcal{D}^*$ converges to a distribution F in the **weak sense** (or in the **sense of distributions**) if $F_n(\varphi) \rightarrow F(\varphi)$ for all $\varphi \in \mathcal{D}$.*

In particular, the following result follows.

Proposition 2.2 (Approximating Distributions with Smooth Functions). *Suppose F is a distribution on $\mathbb{R}^d$. Then there exists a sequence $\{f_n\}$, with $f_n \in C^\infty$, and $f_n \rightarrow F$ in the weak sense.*

However, the space of test functions $C_c^\infty(\mathbb{R}^d)$ could potentially be too restrictive at points. Thus we generalize instead to the Schwartz space, $\mathcal{S} = \mathcal{S}(\mathbb{R}^d)$ of **Schwartz functions**, which are smooth functions that rapidly decrease at infinity with all of their derivatives. I.e., we define

$$\mathcal{S}(\mathbb{R}^d) := \{\varphi \in C_c^\infty(\mathbb{R}^d) : \sup_{x \in \mathbb{R}^d} |x^\beta (\partial_x^\alpha \varphi)(x)| < \infty, \quad \forall |\alpha|, |\beta| \leq N \text{ and } \forall N \in \mathbb{N}\}.$$

We denote,

$$\|\varphi\|_N = \sup_{x \in \mathbb{R}^d, |\alpha|, |\beta| \leq N} |x^\beta (\partial_x^\alpha \varphi)(x)|, \quad \forall N \in \mathbb{N};$$

we say that $\varphi_k \rightarrow \varphi$ in $\mathcal{S}$ iff $\|\varphi_k - \varphi\|_N \rightarrow 0$ for every $N \geq 1$. Accordingly, we make the following definition.

Definition 2.10 (Tempered Distributions). *F is said to be a **tempered distribution** if it is a linear functional on $\mathcal{S}$ which is continuous in the sense that $F(\varphi_k) \rightarrow F(\varphi)$ whenever $\varphi_k \rightarrow \varphi$ in $\mathcal{S}$. By $\mathcal{S}^*$, we shall denote the space of all tempered distributions on a domain.*

Tempered distributions satisfy the following bounding criterion.

Proposition 2.3 (Bounding for Distributions). *Suppose F is a tempered distribution. Then there is a positive integer N and a constant $c > 0$, so that*

$$|F(\varphi)| \leq c \|\varphi\|_N, \quad \forall \varphi \in \mathcal{S}.$$

Proof. Assume that this bound is not satisfied. Then for every positive integer n , there exists $\psi_n \in \mathcal{S}$ with $\|\psi_n\|_n = 1$, and $|F(\psi_n)| \geq n$. For each n , set $\varphi_n = \psi_n/n^{1/2}$. This means $\|\varphi_n\|_N \leq \|\varphi_n\|_n$ for all $n \geq N$. In particular, $\|\varphi_n\|_N \leq n^{-1/2} \rightarrow 0$ as $n \rightarrow \infty$. On the other hand, $|F(\varphi_n)| \geq n^{1/2} \rightarrow \infty$, which contradicts the continuity of F . $\square$

Now we recall that the Fourier transform of Schwartz functions are Schwartz. As a result, it follows that the Fourier transform maps the class of tempered distributions onto itself, so that $\mathcal{S}^*$ is closed under the Fourier transform. But we have not yet described what it means to take the Fourier transform of a distribution.

Recall that for all $\varphi, \psi \in \mathcal{S}$, one has

$$\begin{aligned} \int_{\mathbb{R}^d} \hat{\varphi}(x) \psi(x) dx &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} \varphi(y) \psi(x) e^{-2\pi i y \cdot x} dy \right) dx \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} \varphi(y) \psi(x) e^{-2\pi i y \cdot x} dx \right) dy \quad (\text{by Fubini's Theorem}) \\ &= \int_{\mathbb{R}^d} \varphi(y) \hat{\psi}(y) dy =: \int_{\mathbb{R}^d} \varphi(x) \hat{\psi}(x) dx. \end{aligned}$$

Thus, one seeks to generalize this to distributions as follows:

Definition 2.11 (Fourier Transform for Tempered Distributions). *Let $F \in \mathcal{S}^*$ be a tempered distribution. Then the **Fourier distribution** of F , denoted by $F^\wedge$, is defined by*

$$F^\wedge(\varphi) = F(\varphi^\wedge), \quad \forall \varphi \in \mathcal{S}.$$

In particular, the Fourier transform of tempered distributions satisfy the same properties as the Fourier transform for functions.

2.2.1 Important Examples of Distributions

3 Linear Wave Equation

3.1 Introduction

Through these notes, our goal is to study the properties of the nonlinear wave equation, which can either be a “simple” scalar equation or a system of equations [5]:

$$\frac{1}{\sqrt{-\det(g)}} \partial_\mu \left(\sqrt{-\det(g)} g^{\mu\nu} \partial_\nu \phi \right) = F(\phi, \partial\phi), \quad (3)$$

where g is a Lorentzian metric on $I \times \mathbb{R}^n$, and $\phi : I \times \mathbb{R}^n \rightarrow \mathbb{R}$ is an unknown function of some regularity. For example, if $n = 1$, and g is the flat Minkowski metric, $\text{diag}(1, -1)$, then the nonlinear wave equation is merely

$$\partial_\mu (\partial^{\mu\nu} \partial_\nu \phi) = F(\phi, \partial\phi) \iff \partial_t^2 - (\partial_x \phi)^2 = F(\phi, \partial\phi).$$

Note that we denote the coordinate of I by t (to designate the *time* coordinate), and the coordinates of $\mathbb{R}^n$ by $x = (x^1, \dots, x^n)$. An equation of the form (3) is said to be

- (i) **linear**: g is independent of ϕ and F is a linear function of both of its arguments;
- (ii) **semilinear**: g is independent of ϕ and F is a nonlinear function;
- (iii) **quasilinear**: g is independent of ϕ and/or $\partial\phi$.

Before we build up to the nonlinear wave equation, we will first examine the linear wave equation. Generally speaking, we are interested in the following properties:

- (i) **Existence and Uniqueness**: Does a solution to (3) exist, and if so, is it the only solution?
- (ii) **Dispersion**: What physical properties do solutions to (3) have?
- (iii) **Well-Posedness**: If we consider the *Cauchy wave problem* consisting of the equation (3) along with initial data $(\phi, \partial_t \phi)$ on the hypersurface $\{t = 0\}$, where $\phi, \partial_t \phi$, does there exist a unique solution to the problem that satisfy the initial conditions and is continuously dependent on the initial data?

We will gradually work through each of these results, sometimes developing new techniques and machinery. We first begin by developing some general analytic results, and idea of energy estimates. Then we will apply these to the study of linear wave equations. This will then give a foundation for the study of nonlinear wave equations.

3.2 Energy Estimates

In this section, we introduce the notion of the energy of a solution to the wave equation. We begin by considering the homogeneous linear Cauchy problem,

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1), \end{cases} \quad (4)$$

where we initially assume $\phi_0, \phi_1 \in C_c^\infty \subset \mathcal{S}(\mathbb{R}^n)$, and $\square$ denotes the **wave operator**. As of now, we do not yet have an existence (or uniqueness) result for this Cauchy wave problem. Thus we simply assume that such a unique (sufficiently regular) solution exists for now. Then we define the **energy** of a solution to be the following L^2 -type norm,

Definition 3.1 (Energy). *For $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c(\mathbb{R}^n)$ and ϕ the solution to the Cauchy problem (4), the **energy** of ϕ at the spacetime point (t, x) is defined to be the L^2 -type norm,*

$$E(t) = \frac{1}{2} \int_{\mathbb{R}^n} \left((\partial_t \phi)^2 + \sum_{j=1}^n (\partial_j \phi)^2 \right) (t, x) \, dx. \quad (5)$$

Intuitively, the definition for the energy is as we expected: consider an oscillating 1-dimensional string, and let $\phi(t, x)$ denote the amplitude of the wave at a point x at time t . Then the quantity $(1/2)(\partial_t \phi)^2$ resembles the kinetic energy of an infinitesimal point particle at the spacetime location (t, x) , while the quantity $\sum_{j=1}^n (\partial_j \phi)^2$ resembles the (spring) potential energy at (t, x) . Thus under these observations, $E(t)$ resembles the total energy of the system at some time t . Since the Cauchy wave problem (4) has no source term (i.e., is homogeneous), one expects the following result [5]:

Proposition 3.1 (Conservation of Energy). *Let ϕ be a (sufficiently regular) solution to (4). Define $E(t)$ as given in Definition 3.1. Then E is constant in time.*

Remark 3.1. *We will give two proofs for this result; one relies on straightforward differentiation of $E(t)$ and follows the steps outlined in [5], while the other introduces the concept of conservation laws and follows the steps outlined in [4].*

Proof # 1. Let ϕ be a sufficiently regular solution to (2), and define the energy $E(t)$ of ϕ to be the function,

$$E(t) = \frac{1}{2} \int_{\mathbb{R}^n} \left((\partial_t \phi)^2 + \sum_{j=1}^n (\partial_j \phi)^2 \right) (t, x) \, dx,$$

where we assume that the initial data ϕ_0, ϕ_1 are both smooth with compact support. As we will see in a later section, $\phi \in C^\infty(I, \mathcal{S}(\mathbb{R}^n))$. Thus by the Dominated Convergence Theorem allows us to pass a time derivative inside the integral sign. We then obtain,

$$\begin{aligned} \frac{d}{dt} E(t) &= \frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi)(\partial_t^2 \phi) + \sum_{j=1}^n \partial_t (\partial_j \phi)^2 \, dx \\ &= \frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi) \cdot \left(\sum_{j=1}^n \partial_j^2 \phi \right) + \sum_{j=1}^n \partial_t \phi \cdot (\partial_j \phi)^2 \, dx \\ &= \frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi) \cdot \left(\sum_{j=1}^n \partial_j^2 \phi \right) + \frac{1}{2} \sum_{j=1}^n \int_{\mathbb{R}^n} \partial_j (\partial_t \phi)(\partial_j \phi) \, dx \\ &= \frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi) \cdot \left(\sum_{j=1}^n \partial_j^2 \phi \right) + \frac{1}{2} \sum_{j=1}^n \left[(\partial_t \phi)(\partial_j \phi) \Big|_{|x|=\infty} - \int_{\mathbb{R}^n} (\partial_t \phi)(\partial_j^2 \phi) \, dx \right] \\ &= \frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi) \cdot \left(\sum_{j=1}^n \partial_j^2 \phi \right) \, dx - \frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi) \cdot \left(\sum_{j=1}^n \partial_j^2 \phi \right) \, dx = 0, \end{aligned}$$

where the boundary term is identically zero since ϕ is Schwartz. Thus, the energy is conserved. □

Proof # 2. We start by claiming that solutions to (4) are the stationary points of a variational integral. Define the linear functional $\mathcal{L} : C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n)) \rightarrow \mathbb{R}$ by

$$\mathcal{L}(\phi) = \frac{1}{2} \int_{\mathbb{R} \times \mathbb{R}^n} (\partial_t \phi)^2 - \sum_1^n (\partial_j \phi)^2 dt dx;$$

to obtain this functional, we can use the Euler-Lagrange equations for a physical system. Now let $\psi \in C_c^\infty(\mathbb{R} \times \mathbb{R}^n)$. We recall that $\langle \mathcal{L}'(\phi), \psi \rangle$, where $\mathcal{L}'(\phi)$ is the Fréchet derivative is defined as the directional derivative along ψ ; i.e.,

$$\begin{aligned} \langle \mathcal{L}'(\phi), \psi \rangle &= \left. \frac{d}{d\varepsilon} \mathcal{L}(\phi + \varepsilon \psi) \right|_{\varepsilon=0} \\ &= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \left(\mathcal{L}(\phi) + \varepsilon \int_{\mathbb{R} \times \mathbb{R}^n} (-\nabla \phi \cdot \nabla \psi + (\partial_t \phi)(\partial_t \psi)) dt dx + \mathcal{O}(\varepsilon^2) \right) \\ &= \int_{\mathbb{R} \times \mathbb{R}^n} -(\nabla \phi) \cdot (\nabla \psi) + (\partial_t \phi)(\partial_t \psi) dt dx. \end{aligned}$$

Since ψ is smooth with compact support, and ϕ is Schwartz, using Fubini's theorem and integration by parts, and recognizing that the boundary terms are identically zero, one obtains

$$\langle \mathcal{L}'(\phi), \psi \rangle = - \int_{\mathbb{R} \times \mathbb{R}^n} (\square \phi) \psi dt dx = 0.$$

Thus since $\phi \in C_c^\infty$ was arbitrary, we conclude that ϕ is stationary for $\mathcal{L}$ iff $\square \phi = 0$. Now, we recall Nöther's theorem, which states that if $\mathcal{L}$ is invariant with respect to a 1-parameter family of diffeomorphisms, then there exists a conservation law for an extreme value of $\mathcal{L}$. We note that $\mathcal{L}$ is time-translation invariant: if $G_\varepsilon : \phi(t, x) \mapsto \phi(t + \varepsilon, x)$, then one observes that $\mathcal{L}(\phi) = \mathcal{L}(G_\varepsilon \phi)$. Therefore, since the vector field ∂_t generates the flow corresponding to time translations, one has

$$\begin{aligned} 0 &= \phi_t \square \phi \\ &= \phi_{tt} \phi_t - \Delta \phi \phi_t \\ &= \partial_t \left(\frac{(\partial_t \phi)^2}{2} - \frac{(\nabla \phi)^2}{2} \right) - \operatorname{div}(\nabla \phi \phi_t). \end{aligned}$$

Then, integrating over $\mathbb{R}^n$ yields,

$$0 = \frac{d}{dt} \left(\frac{1}{2} \int_{\mathbb{R}^n} (\partial_t \phi)^2 - (\nabla \phi)^2 dx \right) - \int_{\mathbb{R}^n} \operatorname{div}(\nabla \phi \phi_t) dx.$$

The last term vanishes by using integration by parts and recalling that ϕ is Schwartz. Thus, one has that the energy is independent of time. $\square$

Remark 3.2. As the second proof shows, because of the time translation invariance, to show that the energy is conserved, it suffices to integrate the quantity $(\partial_t \phi)(\square \phi)$ by parts.

Likewise, we have the following conservation law in Sobolev norms [3].

Corollary 3.1 (Sobolev Conservation Law I). *Given initial data $(\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n)$, and assuming the existence of a solution $\phi \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n))$ to the Cauchy wave problem (4), for all $s \in \mathbb{R}$,*

$$\|\partial_t \phi\|_{H^{s-1}(\mathbb{R}^n)} + \|\nabla \phi\|_{H^{s-1}(\mathbb{R}^n)} = \|\phi_0\|_{H^{s-1}(\mathbb{R}^n)} + \|\nabla \phi_1\|_{H^{s-1}(\mathbb{R}^n)}. \quad (6)$$

Proof. Assume that $\phi \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n))$ is a solution to (4) with initial data $(\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n)$. If $\hat{\cdot}$ denotes the Fourier transform of $\cdot$, then one has the equation

$$(\partial_t \phi)^2 + 4\pi^2 |\xi|^2 (\hat{\phi})^2 = (\hat{\phi}_0)^2 + 4\pi^2 |\xi|^2 (\hat{\phi}_1)^2.$$

Note, this equation becomes clearer once we establish a representation formula for ϕ in a later section. Multiplying through by $(1 + |\xi|^2)^{s-1}$ (for arbitrary $s \in \mathbb{R}$), and integrating, yields the desired conservation law. $\square$

One immediate advantage of having conservation laws is that if the Cauchy problem (4) has a solution, then these laws immediately imply uniqueness.

Corollary 3.2 (Uniqueness I). *Given initial data $(\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in H^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)$, if we have two solutions $\phi^{(1)}$ and $\phi^{(2)}$ to the linear wave equation (4) with the given data such that both of them satisfy $(\phi^{(i)}, \partial_t \phi^{(i)}) \in L^\infty([0, T]; H^1(\mathbb{R}^n)) \times L^\infty([0, T]; L^2(\mathbb{R}^n))$ for $i = 1, 2$, then $\phi^{(1)} = \phi^{(2)}$ almost everywhere in $[0, T] \times \mathbb{R}^n$.*

To extract more useful information, we can localize the energy identity, thus obtaining the following result.

Proposition 3.2 (Localized Energy Identity). *Define, for $0 \leq t \leq R$,*

$$E(t; x, R) := \frac{1}{2} \int_{B(x, R-t)} \left((\partial_t \phi)^2 + \sum_1^n (\partial_j \phi)^2 \right) (t, y) dy. \quad (7)$$

Then $E(t; x, R)$ is a nonincreasing function of t .

Remark 3.3. *This result tells us that the energy at time t is less than or equal to the energy at time $s > t$ in the backwards light cone with vertex at x .*

Proof. As suggested in the second proof of Proposition 3.1, we will start with the expression $(\partial_t \phi) \square \phi = 0$ and integrate by parts in the region to the future of $\{t_1\} \times B(x, R-t_1)$ and to the past of $(\{t_2\} \times B(x, R-t_2)) \cup (\cup_{s \in [t_1, t_2]} (\{s\} \times \partial B(x, R-s)))$. The idea behind this choice is that the boundary terms conveniently give us $E(t_1; x, R)$ and $E(t_2; x, R)$. The boundary term on $\cup_{s \in [t_1, t_2]} (\{s\} \times \partial B(x, R-s))$ then has a definite sign, which then allows us to compare the final and initial energy. Indeed, one has the following computations,

$$\begin{aligned} 0 &= \int_{t_1}^{t_2} \int_{B(x, R-t)} \left((\partial_t \phi) \left(-\partial_t^2 \phi + \sum_1^n \partial_j^2 \phi \right) \right) (t, y) dy dt \\ &= - \int_{t_1}^{t_2} \int_{B(x, R-t)} \partial_t \left(\frac{1}{2} (\partial_t \phi)^2 + \frac{1}{2} \sum_1^n (\partial_j \phi)^2 \right) dy dt + \sum_1^n \int_{t_1}^{t_2} \int_{B(x, R-t)} \partial_j ((\partial_t \phi) \cdot (\partial_j \phi)) dy dt \\ &= - \int_{t_1}^{t_2} \left[\frac{1}{2} \frac{d}{dt} \int_{B(x, R-t)} (\partial_t \phi)^2 + \Delta \phi \right] dt - \frac{1}{2} \int_{t_1}^{t_2} \int_{\partial B(x, R-t)} \left((\partial_t \phi)^2 + \sum_1^n (\partial_j \phi)^2 \right) \\ &\quad + \int_{t_1}^{t_2} \int_{\partial B(x, R-t)} (\partial_t \phi) \cdot \sum_1^n (\partial_j \phi) \cdot \frac{(y-x)_j}{|y-x|} d\sigma(y) dt \\ &= -E(t_2; x, R) + E(t_1; x, R) - \frac{1}{2} \int_{t_1}^{t_2} \int_{\partial B(x, R-t)} \left((\partial_t \phi)^2 + \sum_1^n (\partial_j \phi)^2 - 2(\partial_t \phi) \cdot \sum_1^n \frac{(y-x)_j}{|y-x|} (\partial_j \phi) \right) d\sigma(y) dt. \end{aligned}$$

where the second equality follows from Theorem C.4.6 of [1]. Now we estimate the final term of the last equation. Cauchy-Schwarz tells us the following (CS arises in the second line),

$$\begin{aligned} 2(\partial_t \phi) \cdot \sum_1^n \frac{(y-x)_j}{|y-x|} (\partial_j \phi) &\leq (\partial_t \phi)^2 + \left(\sum_1^n \frac{(y-x)_j}{|y-x|} (\partial_j \phi) \right)^2 \\ &\leq (\partial_t \phi)^2 + \left(\sum_1^n \frac{(y-x)_j}{|y-x|} \right)^2 \cdot \left(\sum_1^n (\partial_j \phi)^2 \right) \\ &= (\partial_t \phi)^2 + \sum_1^n (\partial_j \phi)^2. \end{aligned}$$

Thus this implies that the final term is negative. So then, one must have $-E(t_2; x, R) + E(t_1; x, R) \geq 0$, whence $E(t_2; x, R) \leq E(t_1; x, R)$. $\square$

While we are at it, we also consider the following linear, non-constant coefficient wave equation,

$$\begin{cases} \partial_\alpha (a^{\alpha\beta} \partial_\beta \phi) = F, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1), \end{cases} \quad (8)$$

Give proof (possibly using mollifiers). See Holzegel

where a is assumed to be a symmetric $(n+1) \times (n+1)$ matrix on $I \times \mathbb{R}^n$ satisfying

$$\sum_{\alpha, \beta} |a^{\alpha\beta} - m^{\alpha\beta}| < \frac{1}{10}.$$

While the energy is not conserved, in general, due to the source term F , the growth of the energy is controlled in the following sense.

Theorem 3.1 (Growth Bound on Energy). *Assume ϕ is a solution to (8) of some sufficient regularity; let $E(t)$ be the energy of ϕ in the sense of Proposition (3.1). Then there exists a positive constant $C = C(n) > 0$ such that the following energy estimate holds*

$$\|E^{1/2}\|_{C([0, T])} \leq C \left(\|(\phi_0, \phi_1)\|_{\dot{H}^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)} + \int_0^T \|F\|_{L^2(\mathbb{R}^n)}(t) dt \right) \exp \left(C \int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t) dt \right), \quad (9)$$

where by $\partial \cdot$, we denote $(\partial_t \cdot)^2 + \Delta \cdot$.

To prove this result, we first require **Grönwall's lemma**.

Lemma 3.1 (Grönwall's Lemma). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a nonnegative continuous function and $g : \mathbb{R} \rightarrow \mathbb{R}$ be a nonnegative integrable function such that*

$$f(t) \leq A + \int_0^t f(s)g(s) ds$$

holds for some $A \geq 0$, and every $t \in [0, T]$. Then

$$f(t) \leq A \exp \left(\int_0^t g(s) ds \right)$$

for every $t \in [0, T]$.

Proof.

- (1) (Standard Proof): The proof follows by showing that the result holds for any $A > 0$, and then passing to the limit $A \rightarrow 0$. First, we compute

$$\frac{d}{dt} \left(A + \int_0^t f(s)g(s) ds \right) = f(t)g(t) \leq g(t) \left(A + \int_0^t f(s)g(s) ds \right).$$

In particular, this implies that

$$\frac{d}{dr} \log \left[A + \int_0^t f(s)g(s) ds \right] \leq g(r),$$

so that integrating,

$$\log \left[A + \int_0^t f(s)g(s) ds \right] \leq \int_0^t g(r) dr + \log A.$$

Thus, we conclude that

$$f(t) \leq A + \int_0^t f(s)g(s) ds \leq A \exp \left(\int_0^t g(r) dr \right).$$

- (2) (Bootstrap Method): Now we will prove the same result again, using the **bootstrap** method. For each $\varepsilon > 0$, consider the following condition:

$$f(t) \leq (1 + \varepsilon)A \exp \left((1 + \varepsilon) \int_0^t g(s) ds \right), \quad (*)$$

and define the subset $B \subseteq [0, T]$ by

$$B := \{t \in [0, T] : (\star) \text{ holds for all } s \in [0, t]\}.$$

It follows immediately that B is nonempty as it at least contains the point $0 \in B$. Likewise, by continuity of f , it follows that B is closed. By continuity of f , if we can show that f satisfies a *better* estimate for $t \in B$, then it must follow that f satisfies the above bound in some small neighborhood of t , which would show that B is open in $[0, T]$. Indeed, let $t \in B$. Then combining the hypothesis, one has that

$$\begin{aligned} f(t) &\leq A + \int_0^t g(s) ds \leq A + (1 + \varepsilon)A \int_0^t g(s) \exp\left((1 + \varepsilon) \int_0^s g(r) dr\right) ds \\ &\leq A \left(1 + \left(\exp\left((1 + \varepsilon) \int_0^t g(s) ds\right) - 1\right)\right) \quad (\text{by integration by parts}) \\ &= A \exp\left((1 + \varepsilon) \int_0^t g(s) ds\right). \end{aligned}$$

Thus, this shows that B must be open. Since $[0, T]$ is connected, one must have $B = [0, T]$. Hence the modified condition holds for all $t \in [0, T]$ and $\varepsilon > 0$ so that passing to the limit $\varepsilon \rightarrow 0$ yields the desired estimate. □

Now we will use Grönwall's lemma to prove the energy estimate from above.

Proof of Theorem 3.1. For the sake of this proof, we will be assuming that ϕ is compactly supported so that we can ignore the spatial boundary terms in the integration. The fundamental idea behind the techniques in this proof is to bound the energy of the solution on the time interval $[0, T]$ by looking at the total energy at some fixed time, and then summing these contributions. In other words, we wish to integrate the first term of the quantity, $\partial_t \phi (\partial_\alpha (a^{\alpha\beta} \partial_\beta \phi) - F)$ by parts. Before doing so, we make the following observations.

- (1) Suppose first that $\alpha = \beta = 0$. Then one has that

$$\begin{aligned} \partial_t \phi (\partial_\alpha (a^{\alpha\beta} \partial_\beta \phi)) &= (\partial_t \phi) (\partial_t (a^{tt} \partial_t \phi)) \\ &= (\partial_t \phi) ((\partial_t a^{tt}) (\partial_t \phi)) + a^{tt} (\partial_t \phi) (\partial_{tt} \phi) \\ &= (\partial_t a^{tt}) (\partial_t \phi)^2 + \frac{1}{2} a^{tt} \partial_t (\partial_t \phi)^2, \end{aligned}$$

so that

$$\begin{aligned} \int_0^T \int_{\mathbb{R}^n} (\partial_t \phi) (\partial_t (a^{tt} \partial_t \phi)) dx dt &= \int_0^T \int_{\mathbb{R}^n} (\partial_t a^{tt}) (\partial_t \phi)^2 + \frac{1}{2} a^{tt} \partial_t (\partial_t \phi)^2 dx dt \\ &= \frac{1}{2} \int_{\{T\} \times \mathbb{R}^n} a^{tt} (\partial_t \phi)^2 dx - \frac{1}{2} \int_{\{0\} \times \mathbb{R}^n} a^{tt} (\partial_t \phi)^2 dx \\ &\quad + \frac{1}{2} \int_0^T \int_{\mathbb{R}^n} (\partial_t a^{tt}) (\partial_t \phi)^2 dx dt, \end{aligned}$$

where the second line follows from Fubini/Tonelli, and the fundamental theorem of calculus.

- (2) Now assume that one of the indices is t , while the other index runs from $j = 1, \dots, n$. Then one has that

$$\begin{aligned} \partial_t \phi (\partial_\alpha (a^{\alpha\beta} \partial_\beta \phi)) &= (\partial_t \phi) \cdot [\partial_j (a^{jt} \partial_t \phi) + \partial_t (a^{tj} \partial_j \phi)] \\ &= (\partial_t \phi) \cdot [(\partial_j a^{jt}) (\partial_t \phi) + a^{jt} \partial_j (\partial_t \phi) + \partial_t (a^{jt}) (\partial_j \phi) + a^{jt} \partial_j (\partial_t \phi)] \\ &= (\partial_j a^{jt}) (\partial_t \phi)^2 + a^{jt} \partial_j (\partial_t \phi)^2 + (\partial_t a^{jt}) (\partial_t \phi) (\partial_j \phi). \end{aligned}$$

Observe that $a^{jt} = a^{tj}$ by symmetry of a . We first consider the integral of the middle term. Integration by parts tells us that

$$\begin{aligned} \int_0^T \int_{\mathbb{R}^n} a^{jt} \partial_j (\partial_t \phi)^2 dx dt &= \int_0^T \left[a^{jt} (\partial_t \phi)^2 \Big|_{\text{bdr.}} - \int_{\mathbb{R}^n} (\partial_j a^{jt}) (\partial_t \phi)^2 dx \right] dt \\ &= - \int_0^T \int_{\mathbb{R}^n} (\partial_j a^{jt}) (\partial_t \phi)^2 dx dt, \end{aligned}$$

where the boundary terms vanish because of our assumption that ϕ is compactly supported. Thus in fact,

$$\int_0^T \int_{\mathbb{R}^n} \partial_t \phi (\partial_\alpha (a^{\alpha\beta} \partial_\beta \phi)) dx dt = \int_0^T \int_{\mathbb{R}^n} (\partial_j a^{jt}) (\partial_t \phi)^2 - (\partial_j a^{jt}) (\partial_t \phi)^2 + (\partial_t a^{jt}) (\partial_t \phi) (\partial_j \phi) dx dt.$$

(3) Now assume that both indices run over $i, j = 1, \dots, n$. Then one has that

$$\begin{aligned} \int_0^T \int_{\mathbb{R}^n} \partial_t \phi (\partial_i (a^{ij} \partial_j \phi)) dx dt &= (\partial_t \phi) (a^{ij} \partial_j \phi) \Big|_{\text{bdr.}} - \int_0^T \int_{\mathbb{R}^n} \partial_t \partial_i \phi a^{ij} \partial_j \phi dx dt \\ &= -\frac{1}{2} \int_0^T \int_{\mathbb{R}^n} a^{ij} \partial_t (\partial_i \phi \partial_j \phi) dx dt \\ &= -\frac{1}{2} \int_{\{T\} \times \mathbb{R}^n} a^{ij} \partial_i \phi \partial_j \phi dx + \frac{1}{2} \int_{\{0\} \times \mathbb{R}^n} a^{ij} \partial_i \phi \partial_j \phi dx \\ &\quad + \frac{1}{2} \int_0^T \int_{\mathbb{R}^n} (\partial_t a^{ij} (\partial_i \phi) (\partial_j \phi)) dx dt, \end{aligned}$$

where the final line follows from the fact that

$$\partial_t (a^{ij} \partial_i \phi \partial_j \phi) - (\partial_t a^{ij}) (\partial_i \phi) (\partial_j \phi) = a^{ij} \partial_t (\partial_i \phi \partial_j \phi),$$

we used the fundamental theorem of calculus after using the above fact, and the boundary terms in the first line vanish since ϕ has compact support.

Now we obtain the following result:

$$\begin{aligned} \int_0^T \int_{\mathbb{R}^n} (\partial_t \phi) (\partial_\alpha (a^{\alpha\beta} \partial_\beta \phi)) &= \int_0^T \int_{\mathbb{R}^n} (\partial_t \phi) F \\ \implies \frac{1}{2} \int_{\{T\} \times \mathbb{R}^n} a^{tt} (\partial_t \phi)^2 dx - \frac{1}{2} \int_{\{T\} \times \mathbb{R}^n} a^{ij} \partial_i \phi \partial_j \phi dx &= \frac{1}{2} \int_{\{0\} \times \mathbb{R}^n} a^{tt} (\partial_t \phi)^2 dx - \frac{1}{2} \int_{\{0\} \times \mathbb{R}^n} a^{ij} \partial_i \phi \partial_j \phi dx \\ &\quad + \int_0^T \int_{\mathbb{R}^n} (\partial_t \phi) F - \frac{1}{2} (\partial_t a^{tt}) (\partial_t \phi)^2 - (\partial_j a^{jt}) (\partial_t \phi)^2 \\ &\quad + (\partial_j a^{jt}) (\partial_t \phi)^2 - (\partial_t a^{jt}) (\partial_t \phi) (\partial_j \phi) dx dt \end{aligned}$$

Now, one has that

$$\begin{aligned} \left| \int_0^T \int_{\mathbb{R}^n} (\partial_t \phi) F - \frac{1}{2} (\partial_t a^{tt}) (\partial_t \phi)^2 - (\partial_j a^{jt}) (\partial_t \phi)^2 + (\partial_j a^{jt}) (\partial_t \phi)^2 - (\partial_t a^{jt}) (\partial_t \phi) (\partial_j \phi) dx dt \right| \\ \leq C' \int_0^T \|\partial_t \phi\|_{L^2(\mathbb{R}^n)} \|F\|_{L^2(\mathbb{R}^n)}(t) + \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t) \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2(t) dt, \end{aligned}$$

where we used Hölder's inequality to obtain the first term, and assumed $\partial a \in L^\infty(\mathbb{R}^n)$. On the other hand, by our hypothesis on $a^{\alpha\beta}$, there exists some constant $K > 0$ large enough so that (taking $C = \min\{C', K\}$),

$$\|\partial \phi\|_{L^2(\mathbb{R}^n)}(T) \leq C \left| \frac{1}{2} \int_{\{T\} \times \mathbb{R}^n} a^{tt} (\partial_t \phi)^2 dx - \frac{1}{2} \int_{\{T\} \times \mathbb{R}^n} a^{ij} \partial_i \phi \partial_j \phi dx \right|.$$

Thus this implies that

$$\|\partial \phi\|_{L^2(\mathbb{R}^n)}(T) \leq C \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2(0) + C \int_0^T \|\partial_t \phi\|_{L^2(\mathbb{R}^n)} \|F\|_{L^2(\mathbb{R}^n)}(t) + \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t) \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2(t) dt.$$

Explain how?

Since one can repeat this argument for any smaller interval $[0, T']$, where $0 < T' < T$, one obtains

$$\sup_{t \in [0, T]} \|\partial \phi\|_{L^2(\mathbb{R}^n)}(t) \leq C \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2(0) + C \int_0^T \|\partial_t \phi\|_{L^2(\mathbb{R}^n)} \|F\|_{L^2(\mathbb{R}^n)}(t) + \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t) \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2(t) dt.$$

For the first term inside the integral, one has that

$$\begin{aligned} \int_0^T \|\partial_t \phi\|_{L^2(\mathbb{R}^n)} \|F\|_{L^2(\mathbb{R}^n)}(t) dt &\leq \sup_{t \in [0, T]} \|\partial_t \phi\|_{L^2(\mathbb{R}^n)} \cdot \left(\int_0^T \|F\|_{L^2(\mathbb{R}^n)}(t) dt \right) \\ &\leq \delta \|\partial_t \phi\|_{L^2(\mathbb{R}^n)}^2 + \frac{1}{4\delta} \left(\int_0^T \|F\|_{L^2(\mathbb{R}^n)}(t) dt \right)^2. \end{aligned}$$

where the second line follows from Young's inequality with ε^3 . If we fix $\delta > 0$ small enough so that the first term can be combined with the left side of the above inequality (and absorbing δ into the constant C), one has that

$$\sup_{t \in [0, T]} \|\partial \phi\|_{L^2(\mathbb{R}^n)}(t) \leq C \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2(0) + C \left(\int_0^T \|F\|_{L^2(\mathbb{R}^n)}^2(t) dt \right)^2 + C \int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t) \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2(t) dt.$$

Thus using Grönwall's lemma concludes the proof. $\square$

We can, of course, say more. Using the above estimate, we can control the size of the higher-order derivatives of ϕ via the following result.

Proposition 3.3 (Higher Order Energy Estimate I). *Assume ϕ is a smooth solution to the Cauchy wave problem 8, and let k be a positive integer. Then there exists some constant $C = C(n, k) > 0$, the following energy estimates holds:*

$$\begin{aligned} &\sup_{t \in [0, T]} \|\phi\|_{H^k(\mathbb{R}^n)}(t) + \sup_{t \in [0, T]} \|\partial_t \phi\|_{H^{k-1}(\mathbb{R}^n)}(t) \\ &\leq C(1 + T) \exp \left(C \int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t) dt \right) \left(\|f\|_{H^k(\mathbb{R}^n)} + \|g\|_{H^{k-1}(\mathbb{R}^n)} + \int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)}(t) dt \right. \\ &\quad \left. + C \int_0^T \sum_{|\alpha|+|\beta| \leq k-1} \|\partial \partial_x^\alpha a \partial \partial_x^\beta \phi\|_{L^2(\mathbb{R}^n)}(t) + \sum_{|\alpha|+|\beta| \leq k-1} \|\partial_x^\alpha a \partial \partial_x^\beta \phi\|_{L^2(\mathbb{R}^n)}(t) dt \right). \end{aligned} \quad (10)$$

Proof. [! Complete Later !] $\square$

³For any $a, b \geq 0$ and any $\varepsilon > 0$,

$$ab \leq \frac{\varepsilon^p a^p}{p} + \frac{1}{\varepsilon^q q} b^q$$

, where p, q are conjugate exponents.

3.3 Solutions to the Linear Wave Equation (Constant Coefficients)

In the previous subsection, we outlined various simple energy estimates solutions to the (non-)constant coefficient linear wave equations would satisfy if they existed, and also used these estimates to establish uniqueness results. But we have not yet shown that such solutions exist, nor what their physical properties. We dedicate this subsection to answering these questions.

Consider again the Cauchy wave problem (4),

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1), \end{cases}$$

where we initially assume $\phi_0, \phi_1 \in C_c^\infty \subset \mathcal{S}(\mathbb{R}^n)$, and $\square$ denotes the **wave operator**. Then one has the following result.

Theorem 3.2 (Existence and Uniqueness for (4)). *Given initial data $(\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n)$, there exists a unique solution $\phi \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n))$ that solves (4).*

Proof. Assuming that ϕ is an appropriate space of functions, taking the Fourier transformation of the wave equation yields,

$$\partial_t^2 \hat{\phi} + 4\pi^2 |\xi|^2 \hat{\phi} = 0 \implies \partial_t^2 \hat{\phi} = -4\pi^2 |\xi|^2 \hat{\phi},$$

which is a second-order ordinary differential equation. Thus solving this system, one obtains the general solution

$$\hat{\phi}(t, \xi) = A \cos(2\pi t |\xi|) + B \sin(2\pi t |\xi|).$$

Thus matching the initial conditions, it follows that

$$\hat{\phi}(t, \xi) = \hat{\phi}_0 \cos(2\pi t |\xi|) + \frac{\hat{\phi}_1}{2\pi |\xi|} \sin(2\pi t |\xi|).$$

But to complete the proof of existence, it remains to be seen that we can compute the inverse Fourier transform of $\hat{\phi}$. It suffices to show, at least, that $\hat{\phi}(t, \cdot) \in L^1(\mathbb{R}^n)$. Thus, we compute,

$$\int_{\mathbb{R}^n} |\hat{\phi}(t, \xi)| d\xi \leq \int_{\mathbb{R}^n} |\hat{\phi}_0(\xi)| d\xi + \int_{\mathbb{R}^n} \frac{|\hat{\phi}_1(\xi)|}{2\pi |\xi|} |\sin(2\pi t |\xi|)| d\xi.$$

Since $\hat{\phi}_0(\xi)$ is Schwartz, it is integrable, and thus the first term is finite. $\hat{\phi}_1(\xi)$ is also Schwartz, and thus bounded above; i.e., there exists some $M > 0$ such that for all ξ , $|\hat{\phi}_1(\xi)| \leq M$. We note that $\sin(2\pi t |\xi|)/|\xi| \rightarrow t$ as $\xi \rightarrow 0$. This means there exists some $\varepsilon > 0$ such that for all $\xi \in B_\varepsilon(0)$, $|\sin(2\pi t |\xi|)/|\xi| - t| \lesssim_t 1$. This means that

$$\begin{aligned} \int_{\mathbb{R}^n} |\hat{\phi}(t, \xi)| d\xi &\leq \int_{\mathbb{R}^n} |\hat{\phi}_0(\xi)| d\xi + \int_{\mathbb{R}^n} \frac{|\hat{\phi}_1(\xi)|}{2\pi |\xi|} |\sin(2\pi t |\xi|)| d\xi \\ &\lesssim \int_{\mathbb{R}^n} |\hat{\phi}_0| d\xi + \int_{B_\varepsilon(0)} M d\xi + \int_{\mathbb{R}^n \setminus B_\varepsilon(0)} |\hat{\phi}_1(\xi)| d\xi < \infty. \end{aligned}$$

Therefore, it follows that $\hat{\phi}(t, \cdot) \in L^1(\mathbb{R}^n)$. The Fourier Inversion Theorem then tells us that

$$\phi(t, x) = \int_{\mathbb{R}^n} \hat{\phi}(t, \xi) e^{2\pi i x \cdot \xi} d\xi$$

The final piece of the proof of existence is that $\phi(t, x)$ is a solution to (4). We first compute,

$$\phi(0, x) = \int_{\mathbb{R}^n} \hat{\phi}(0, \xi) e^{2\pi i x \cdot \xi} d\xi = \int_{\mathbb{R}^n} \hat{\phi}_0 e^{2\pi i x \cdot \xi} d\xi = \phi_0(x).$$

Then using integrability and the dominated convergence theorem, one obtains:

$$\begin{aligned}\partial_t \phi(t, x) &= \int_{\mathbb{R}^n} \partial_t \hat{\phi}(t, \xi) e^{2\pi i x \cdot \xi} d\xi, \\ \partial_t^2 \phi(t, x) &= \int_{\mathbb{R}^n} \partial_t^2 \hat{\phi}(t, \xi) e^{2\pi i x \cdot \xi} d\xi, \\ \Delta \phi(t, x) &= - \int_{\mathbb{R}^n} 4\pi^2 |\xi|^2 \hat{\phi}(t, \xi) e^{2\pi i x \cdot \xi} d\xi,\end{aligned}$$

so that

$$\partial_t \phi(0, x) = \int_{\mathbb{R}^n} \hat{\phi}_1(\xi) e^{2\pi i x \cdot \xi} d\xi = \phi_1(x),$$

and

$$\square \phi = \partial_t^2 \phi(t, x) - \Delta \phi(t, x) = \int_{\mathbb{R}^n} \partial_t^2 \hat{\phi}(t, \xi) + 4\pi^2 |\xi|^2 \hat{\phi}(t, \xi) d\xi = 0.$$

Thus, an integrable solution exists for (4) under the hypothesis assumed above. In fact, since $\hat{\phi}(t, x) \in \mathcal{S}(\mathbb{R}^n)$, one actually has that $\phi(t, \cdot) \in \mathcal{S}(\mathbb{R}^n)$. To prove uniqueness, assume $\phi^{(1)}$ and $\phi^{(2)}$ are two solutions to (4). Linearity of $\square$ then implies that $\phi := \phi^{(1)} - \phi^{(2)}$ also solves the linear wave equation with zero initial data. Thus, if $E(t) = \|\partial \phi\|_{L^2(\mathbb{R}^n)}^2$ denotes the energy of ϕ , then $E(0) = 0$. Thus conservation of energy implies that $E(t) = 0$ for all $t \geq 0$, whence $\phi \equiv 0$. Therefore, $\phi^{(1)} \equiv \phi^{(2)}$. $\square$

In particular, one does not completely require the initial data to smooth with compact support. We have the following existence and uniqueness result.

Theorem 3.3 (Existence and Uniqueness for (4)). *Given initial data $(\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in H^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)$, there exists a unique solution $\phi \in L^\infty([0, T]; H^1(\mathbb{R}^n)) \times L^\infty([0, T]; L^2(\mathbb{R}^n))$ that solves (4).*

Proof. The following proof is due to [3]. [! Complete Later !] $\square$

3.3.1 Fundamental Solutions

Now we will present another approach to solving (4). In particular, we consider instead the *fundamental solution* using distributions (see Section 2.2 for a review of distributions). We immediately have the following definition.

Definition 3.2 (Forward Fundamental Solution). *The distribution E_+ is said to be a **forward fundamental solution** to the linear wave equation on $\mathbb{R}^{n+1}$ if*

1. $\square E_+ = \delta_0$ (where δ_0 is defined as $\delta_0(\varphi) = \varphi(t=0, x=0)$),
2. $\text{supp}(E_+) \subseteq \{(t, x) \in \mathbb{R}^{1+n} \mid 0 \leq |x| \leq t\}$.

In other words, the forward fundamental solution to the linear wave equation is the solution to linear wave equation on the *forward light cone*. Immediately, one has the following uniqueness result.

Proposition 3.4 (Uniqueness of Forward Fundamental Solution). *A forward fundamental solution on $\mathbb{R}^{n+1}$, if it exists, is unique.*

Proof. Let E_1, E_2 be two forward fundamental solutions. Then one has,

$$E_1 = E_1 * \delta_0 = E_1 * (\square E_2) = E_1 * (\square \delta_0 * E_2) = (\square \delta_0 * E_1) * E_2 = \square E_1 * E_2 = \delta_0 * E_2 = E_2,$$

where we use the support properties of E_1 and E_2 to justify well-definedness of the convolution of the distributions and to justify swapping the wave operator. $\square$

Now, we can use the forward fundamental solution to craft a solution to (4).

Proposition 3.5 (Forward Fundamental Solutions and Representation Formulas). *Let E_+ be a forward fundamental solution to the linear wave equation on $\mathbb{R}^{1+n}$. Then if $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n)$, the unique solution to (4) is given by,*

$$\phi(t, x) = -E_+ * (\phi_1 \delta_{\{t=0\}}) - (\partial_t E_+) * (\phi_0 \delta_{\{t=0\}}).$$

More generally, for $\square \phi = F$, where $F \in C^\infty(\mathbb{R}^n)$, ϕ is given for $t \geq 0$ by the following formula

$$\phi(t, x) = -E_+ * (\phi_1 \delta_{\{t=0\}}) - (\partial_t E_+) * (\phi_0 \delta_{\{t=0\}}) + (F \chi_{\{t \geq 0\}}) * E_+,$$

where

$$\chi_{\{t \geq 0\}}(t, x) := \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

is the characteristic function for the set $\{t \geq 0\}$.

Proposition 3.6 (Unique Fundamental Solution). *The unique forward fundamental solution is given by*

$$E_+(t, x) = -\frac{\pi^{\frac{1-n}{2}}}{2} \chi_{\{t \geq 0\}} \chi_+^{-\frac{n-1}{2}}(t^2 - |x|^2),$$

where

$$\chi_+^a := \frac{x^a \chi_{\{x \geq 0\}}}{\Gamma(a+1)}.$$

From the fundamental solution, one has the following result.

Corollary 3.3 (Propagation of Waves). *Solutions to the wave equation $\square \phi = F$ propagate in time; i.e., $\frac{d}{dt} \text{supp}(\phi)(t) > 0$.*

Now we can derive some of the explicit representation formulas given in the Appendix.

Proposition 3.7 (d'Alembert's Formula). *Let $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n)$ be initial data to (4) in $\mathbb{R}^{1+1}$. Then the unique solution is given by*

$$\phi(t, x) = \frac{1}{2} (\phi_0(x+t) - \phi_0(x-t)) + \frac{1}{2} \int_{x-t}^{x+t} \phi_1(y) dy.$$

Proof. First, we have

$$\chi_+^0(t^2 - |x|^2) = \chi_{\{t^2 - |x|^2 \geq 0\}},$$

so that

$$\begin{aligned} (\chi_{\{t \geq 0\}} \chi_+^0(t^2 - |x|^2)) * (\phi_1 \delta_{\{t=0\}}) &= \int_{-\infty}^{\infty} \chi_{\{t \geq 0\}} \chi_{\{t^2 - |x-y|^2\}} \phi_1(y) dy = \int_{x-t}^{x+t} \phi_1(y) dy. \\ (\partial_t (\chi_{\{t \geq 0\}} \chi_+^0(t^2 - |x|^2))) * (\phi_0 \delta_{\{t=0\}}) &= \partial_t \int_{x-t}^{x+t} \phi_0(y) dy = \phi_0(x+t) - \phi_0(x-t). \end{aligned}$$

Thus using the formula for the forward fundamental solution, we conclude that

$$\phi(t, x) = \frac{1}{2} (\phi_0(x+t) - \phi_0(x-t)) + \frac{1}{2} \int_{x-t}^{x+t} \phi_1(y) dy.$$

□

The support of the forward fundamental solution also yields the following results.

Corollary 3.4 (Finite Speed of Propagation). *Suppose $\phi \in C^\infty(\mathbb{R}^{n+1})$ is a solution to the linear wave equation with data (ϕ_0, ϕ_1) . If $(\phi_0, \phi_1) = (0, 0)$ in $\{y \in \mathbb{R}^n : |x-y| \leq t\}$, then $\phi(t, x) = 0$.*

Corollary 3.5 (Strong Huygens Principle). *Let $n \geq 3$ be an odd integer. Suppose $\phi \in C^\infty(\mathbb{R}^{n+1})$ is a solution to the linear wave equation with data (ϕ_0, ϕ_1) . If $(\phi_0, \phi_1) = (0, 0)$ in $\{y \in \mathbb{R}^n : |x-y| = t\}$, then $\phi(t, x) = 0$.*

3.3.2 Decay Estimates

Corollary 3.3 shows that as t increases, the support of the unique solution to the (non)homogeneous wave equation increases (e.g., consider ripples on the surface of a lake). For $n \geq 2$, the measure of the support increases in time, while for $n = 1$, the measure remains constant. However, we showed that the energy of a solution to (4) is conserved. This must imply that the amplitude of the wave $|\phi(t, x)|$ decays as $t \rightarrow \infty$. We have the following decay estimates.

Proposition 3.8 (Pointwise Decay Rate). *Fix any $R > 0$. For $(\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty(\mathbb{R}^n)$ and ϕ the unique solution to the linear Cauchy wave problem (4),*

$$\sup_{|x| \leq R} |\phi(t, x)| \lesssim_{(\phi_0, \phi_1, R)} \frac{1}{(1+t)^{n-1}}. \quad (11)$$

Proof. The fundamental ideas behind the proof are as follows: we wish to consider the expression

$$\phi(t, x) = \int_{\mathbb{R}^n} e^{2\pi i(t|\xi|+x \cdot \xi)} \left(\frac{\hat{\phi}_0}{2} + \frac{\hat{\phi}_1}{2\pi i|\xi|} \right) + e^{-2\pi i t|\xi|} \left(\frac{\hat{\phi}_0}{2} - \frac{\hat{\phi}_1}{2\pi i|\xi|} \right) d\xi,$$

and then bound each of the terms in the integral. For sake of demonstrating the main ideas, we shall assume $n = 3$. The proof follows trivially for all $|t| \leq 1$: let

$$\begin{aligned} I_1 &= \int_{\mathbb{R}^3} e^{2\pi i(t|\xi|+x \cdot \xi)} \frac{\hat{\phi}_0}{2} d\xi, & I_2 &= \int_{\mathbb{R}^3} e^{-2\pi i t|\xi|} \frac{\hat{\phi}_0}{2} d\xi, \\ I_3 &= \int_{\mathbb{R}^3} e^{2\pi i(t|\xi|+x \cdot \xi)} \frac{\hat{\phi}_1}{2\pi i|\xi|} d\xi & I_4 &= - \int_{\mathbb{R}^3} e^{-2\pi i t|\xi|} \frac{\hat{\phi}_1}{2\pi i|\xi|} d\xi. \end{aligned}$$

Since $\hat{\phi}_0$ is Schwartz, and thus integrable, there exist $M_1, M_2 > 0$ such that for all $|t| \leq 1$, $|I_i| \leq M_i$ for $i = 1, 2$. Now for I_3 (and analogously for I_4), we can write

$$\begin{aligned} |I_3| &\leq \int_{\mathbb{R}^3} \frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|} d\xi = \int_{B_1(0)} \frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|} d\xi + \int_{\mathbb{R}^3 \setminus B_1(0)} \frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|} d\xi \\ &\leq \int_{B_1(0)} \frac{K_1}{2\pi|\xi|} d\xi + \int_{\mathbb{R}^3 \setminus B_1(0)} \frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|} d\xi \\ &= \int_{B_1(0)} \frac{K_1}{2\pi|\xi|} d\xi + \int_{\mathbb{R}^3 \setminus B_1(0)} \frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|} d\xi < \infty, \end{aligned}$$

where the second line follows from the fact that since $\hat{\phi}_1 \in \mathcal{S}(\mathbb{R}^n)$, it is continuous on the compact set $\overline{B_1(0)}$ and thus bounded above by some positive constant K_1 , and the final line follows from the fact that $1/|\xi|$ is integrable over $B_1(0)$ for all $n \geq 2$, and the second term in the final line is controlled above by Hölder's inequality; I_4 is controlled in the same way. Thus, one has that for every $x \in \mathbb{R}^n$, and all $|t| \leq 1$, $|\phi(t, x)| \leq M$ for some constant M . Taking $C > 0$ large enough so that $C \geq M \cdot 2^{n-1}$, one has that

$$\sup_{|x| \leq R} |\phi(t, x)| \leq \frac{M}{(1+t)^2};$$

note that since M_1, M_2, K_1, K_2 are dependent on ϕ_0, ϕ_1 , the proof concludes. So now we focus on the case with $t > 1$. Now introduce polar coordinates on $\mathbb{R}^3$ as follows: let $(|\xi|, \xi_\theta, \xi_\phi)$ be the polar coordinates on $\mathbb{R}_\xi^3$ with $\xi_1 = |\xi| \sin \xi_\theta \cos \xi_\phi$, $\xi_2 = |\xi| \sin \xi_\theta \sin \xi_\phi$, and $\xi_3 = |\xi| \cos \xi_\theta$. The corresponding volume form is,

$$d\xi = |\xi|^2 \sin \xi_\theta d|\xi| d\xi_\theta d\xi_\phi.$$

Furthermore, one has that for any $N \in \mathbb{N}_0$,

$$\left(\frac{1}{2\pi i t} \frac{\partial}{\partial |\xi|} \right)^N e^{2\pi i t|\xi|} = e^{2\pi i t|\xi|}.$$

Now using integration by parts,

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^\pi \int_0^\infty e^{2\pi i t |\xi|} e^{2\pi i (x \cdot \xi)} \frac{\hat{\phi}_1(\xi)}{2\pi i} |\xi| \sin(\xi_\theta) d|\xi| d\xi_\theta d\xi_\varphi \\ &= \left[\frac{1}{(2\pi i)^2 t} \int_0^{2\pi} \int_0^\pi e^{2\pi i t |\xi|} e^{2\pi i (x \cdot \xi)} \hat{\phi}_1(\xi) |\xi| \sin(\xi_\theta) d\xi_\varphi d\xi_\theta \right]_{|\xi|=0}^{R \rightarrow \infty} \\ &\quad - \int_0^{2\pi} \int_0^\pi \int_0^\infty \frac{1}{2\pi i t} e^{2\pi i t |\xi|} \frac{\partial}{\partial |\xi|} \left(e^{2\pi i x \cdot \xi} \frac{\hat{\phi}_1(\xi)}{2\pi i} |\xi| \right) \sin(\xi_\theta) d|\xi| d\xi_\theta d\xi_\varphi \end{aligned}$$

We claim that the first term is identically zero. To prove this, we first note that

$$\lim_{|\xi| \rightarrow \infty} \left| e^{2\pi i t |\xi|} e^{2\pi i (x \cdot \xi)} \hat{\phi}_1(\xi) |\xi| \sin(\xi_\theta) \right| = \lim_{|\xi| \rightarrow \infty} |\hat{\phi}_1(\xi)| |\xi| = 0,$$

since $\hat{\phi}_1(\xi) \in \mathcal{S}(\mathbb{R}^3)$. On the other hand, the integrand vanishes identically at $|\xi| = 0$. Now, we see that one application of integration by parts gave us a power of t . To get another power of t , we will use integration by parts on the second term. This time, after dealing with a boundary term at $|\xi| = 0$, we get

$$I = - \int_{\mathbb{R}^3} e^{2\pi i t |\xi|} \frac{1}{8\pi^3 i t^2} (\dots) \sin(\xi_\theta) d|\xi| d\xi_\theta d\xi_\varphi + \frac{1}{2\pi^2 i t^2} \hat{\phi}_1(\xi = 0).$$

The boundary term does not allow us to use integration by parts again. Thus bounding each of the term and using the fact that $\hat{\phi}_1 \in \mathcal{S}$, one obtains

$$|I| \leq \frac{C(1 + R^2)}{t^2}.$$

□

Note that this decay bound is not uniform for all $x \in \mathbb{R}^n$. We can actually get a uniform decay bound for all $x \in \mathbb{R}^n$, but at the cost of the strength.

Theorem 3.4 (Uniform Decay Rate). *For $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n)$, and ϕ the unique solution to the homogeneous initial value problem, there exists $C = C(\phi_0, \phi_1) > 0$ so that*

$$\sup_{x \in \mathbb{R}^n} |\phi(t, x)| \leq \frac{C}{(1 + t)^{\frac{n-1}{2}}}$$

for $t \geq 0$.

Proof. As before, we only present the proof for $n = 3$ and for the term

$$I = \left(\int_{\mathbb{R}^2} e^{2\pi i (t|\xi| + x \cdot \xi)} \frac{\hat{\phi}_1(\xi)}{2\pi i |\xi|} d\xi \right).$$

It is always possible to find a coordinate system in which $x = (0, 0, \alpha)$ for some $\alpha \in \mathbb{R}$; thus, we will work in these coordinates. Let $\rho = \sqrt{\xi_1^2 + \xi_2^2}$, and let us use the coordinate system $(\xi_3, \rho, \xi_\theta)$, where ξ_θ is given by $\rho \cos(\xi_\theta) = \xi_1$. First note that

$$\frac{\partial |\xi|}{\partial \rho} = \frac{\partial}{\partial \rho} \sqrt{\rho^2 + \xi_3^2} = \frac{\rho}{|\xi|}.$$

Moreover, $\partial_\rho e^{2\pi i x \cdot \xi} = 0$. Thus, since for any nonnegative integer N ,

$$\left(\frac{|\xi|}{2\pi i \rho t} \frac{\partial}{\partial \rho} e^{2\pi i t |\xi|} \right)^N = e^{2\pi i t |\xi|},$$

integration by parts gets us,

$$\begin{aligned} I &= \int_{\mathbb{R}^3} \frac{|\xi|}{2\pi i \rho t} \frac{\partial}{\partial \rho} (e^{2\pi i t |\xi|}) e^{2\pi i x \cdot \xi} \frac{\hat{\phi}_1(\xi)}{2\pi i |\xi|} \rho d\rho d\xi_\theta d\xi_3 = \int_{\mathbb{R}^3} \frac{1}{2\pi i t} \frac{\partial}{\partial \rho} (e^{2\pi i t |\xi|}) e^{2\pi i x \cdot \xi} \frac{\hat{\phi}_1(\xi)}{2\pi i} d\rho d\xi_\theta d\xi_3 \\ &= - \int \frac{1}{4\pi^2 t} e^{2\pi i (t|\xi| + x \cdot \xi)} \hat{\phi}_1(\xi) \Big|_{\rho=0}^{\infty} d\xi_\theta d\xi_3 + \int_{\mathbb{R}^3} \frac{1}{4\pi^2 t} e^{2\pi i (t|\xi| + x \cdot \xi)} \frac{\partial}{\partial \rho} (\hat{\phi}_1(\xi)) d\rho d\xi_\theta d\xi_3 \\ &= \int \frac{1}{2\pi t} e^{2\pi i (t|\xi_3| + \alpha \xi_3)} \hat{\phi}_1(\rho=0, \xi=\xi_3) d\xi_3 + \int_{\mathbb{R}^3} \frac{1}{4\pi^2 t} e^{2\pi i (t|\xi| + x \cdot \xi)} \frac{\partial}{\partial \rho} (\hat{\phi}_1(\xi)) d\rho d\xi_\theta d\xi_3. \end{aligned}$$

Since $\hat{\phi}_1(\xi)$ and $\partial_\rho \hat{\phi}_1(\xi)$ are Schwartz, it follows that

$$|I| \leq \frac{\|\hat{\phi}_1\|_{L^1}}{2\pi t} + \frac{\|\partial_\rho \hat{\phi}_1\|_{L^1}}{4\pi^2 t} =: \frac{M}{t}.$$

Now we split into two cases. First assume $t \in [0, 1]$. Then $(1+t)^{-1} \geq 2^{-1}$ so that $2/(1+t) \geq 1$. Directly bounding $|I(t, x)|$, one has that $|I(t, x)| \leq M_0$ for some $M_0 > 0$, and all $t \in [0, 1]$, $x \in \mathbb{R}^3$. Thus in this case, $|I| \leq M_0 \leq \frac{2M_0}{1+t}$. Now for the $t > 1$ case, $2t \geq 1+t$ so that $2/(1+t) \geq \frac{1}{t}$. Therefore, $|I| \leq \frac{2M}{1+t}$. Take $C = \max\{2M_0, 2M\}$ so that $|I| \leq C/(1+t)$ for all $t \in [0, 1]$ and $x \in \mathbb{R}^3$. The other terms of $\phi(t, x)$ can be controlled the same way. $\square$

The key result is the following: for all $n \geq 2$, solutions to (4) undergo *geometric dispersion*, while no dispersion occurs for $n = 1$.

3.4 Solutions to the Linear Wave Equation (Variable Coefficients)

In this subsection, we continue studying the (non)homogeneous linear wave equation with *variable* coefficients [5]. Consider again the equation (8),

$$\begin{cases} \partial_\alpha (a^{\alpha\beta} \partial_\beta \phi) = F \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n), \end{cases}$$

where $k \in \mathbb{N}$, $\phi : I \times \mathbb{R}^n \rightarrow \mathbb{R}$, and $a^{\alpha\beta}$ is assumed to be a symmetric matrix such that⁴.

$$\sum_{\alpha, \beta} |a^{\alpha\beta} - m^{\alpha\beta}| < \frac{1}{10}.$$

Furthermore, we shall assume that a and all its derivatives (of all orders) are bounded in $[0, T] \times \mathbb{R}^n$, and that $F : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R} \in L^1([0, T]; H^{k-1}(\mathbb{R}^n))$. As in the case of the linear wave equation with constant coefficients, we wish to proceed with an existence and uniqueness result. But this requires us to first present the following result.

Lemma 3.2 (Bound on Sobolev Norms). *Let L denote the partial differential operator $L\phi = \partial_\alpha (a^{\alpha\beta} \partial_\beta \phi)$. Let L^* be defined by $L^*\psi = \partial_\alpha (a^{\alpha\beta} \partial_\beta \psi)$ be the (formal) adjoint of L . Suppose $\psi \in C_c^\infty((-\infty, T) \times \mathbb{R}^n)$. Then for every $m \in \mathbb{Z}$, there exists $C = C(m, T, a, b) > 0$ such that*

$$\|\psi\|_{H^m(\mathbb{R}^n)}(t) \leq C \cdot \int_t^T \|L^* \psi\|_{H^{m-1}(\mathbb{R}^n)}(s) ds. \quad (12)$$

Proof. **[!! Complete Later !!]** $\square$

Theorem 3.5 (Existence of Solutions to the Linear Equation). *Let $k \in \mathbb{N}$. Given $F \in L^1([0, T]; H^{k-1}(\mathbb{R}^n))$, there exists a solution*

$$(\phi, \partial_t \phi) \in L^\infty([0, T]; H^k(\mathbb{R}^n)) \times L^\infty([0, T]; H^{k-1}(\mathbb{R}^n))$$

solving (8).

⁴to ensure hyperbolicity; **[!! Complete Later !!]**

Explain?

Proof. Unlike for the linear case with constant coefficients where we explicitly derived a representation formula, the proof of this result requires a bit of functional analysis (in particular the *Hahn-Banach theorem*). The proof will be carried out over two steps.

1. First assume that $(\phi_0, \phi_1) = (0, 0)$. Let L be the differential operator $\partial_\alpha(a^{\alpha\beta}\partial_\beta\bullet)$, and let $L^* : C_c^\infty((-\infty, T) \times \mathbb{R}^n) \rightarrow C_c^\infty((-\infty, T) \times \mathbb{R}^n)$ be the formal adjoint of L ⁵. For every $\psi \in C_c^\infty((-\infty, T) \times \mathbb{R}^n)$, define a map

$$\Phi : L^* \psi \in L^\infty(C_c^\infty((-\infty, T) \times \mathbb{R}^n)) \mapsto \int_0^T \int_{\mathbb{R}^n} \psi F \, dx dt =: \langle F, \psi \rangle.$$

We claim that this functional Φ is bounded. Indeed, one has

$$\begin{aligned} |\Phi(L^* \psi)| &= \left| \int_0^T \int_{\mathbb{R}^n} \psi F \, dx dt \right| \leq \int_0^T \int_{\mathbb{R}^n} |\psi F| \, dx dt \\ &\leq \int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)} \|\psi\|_{H^{1-k}(\mathbb{R}^n)} \, dt \\ &\leq C \cdot \left(\int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)} \, dt \right) \cdot \left(\sup_{t \in [0, T]} \|\psi\|_{H^{1-k}(\mathbb{R}^n)} \right) \\ &\leq C' \cdot \int_0^T \|L^* \psi\|_{H^{-k}(\mathbb{R}^n)}(s) \, ds, \end{aligned}$$

where (1) the second line follows from the generalized Hölder's inequality⁶, (2) the final line follows from incorporating $\int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)} \, dt$ into C , and using the previous lemma. It is also straightforward to see that the Φ is linear. Thus, by the Hahn-Banach theorem, there exists $\phi \in (L^1((-\infty, T); H^{-k}(\mathbb{R}^n)))^* = L^\infty((-\infty, T); H^k(\mathbb{R}^n))$ with $\phi = 0$ for $t < 0$ that extends Φ : $\langle F, \psi \rangle = \langle \phi, L^* \psi \rangle$ for every $\psi \in C_c^\infty((-\infty, T) \times \mathbb{R}^n)$. Thus ϕ is a *weak solution* (see e.g., Section 2.2) to the Cauchy wave problem with trivial data.

2. Now turn to the general case with $(\phi_0, \phi_1) \in H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n)$, and let u be a function in $[0, T] \times \mathbb{R}^n$ such that

$$(u, \partial_t u) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1).$$

Then the system $L\eta = F - Lu$ has trivial initial data. Then the work in (1) gives us a weak solution η to the problem. Setting $\phi = \eta + u$ gives the desired solution⁷.

□

Remark 3.4. *It is important to note that the above theorem only gives the existence of weak solutions. If the initial data is sufficiently regular, we can upgrade to a classical solution via the Sobolev Embedding Theorem.*

Theorem 3.6 (Sobolev Embedding Theorem). *For every $s > \frac{n}{2}$, and for all $\phi \in H^s(\mathbb{R}^n)$,*

$$\|\phi\|_{L^\infty(\mathbb{R}^n)} \lesssim_{(n,s)} \|\phi\|_{H^s(\mathbb{R}^n)}.$$

I.e., for every $s > n/2$, $H^s(\mathbb{R}^n) \hookrightarrow C_0(\mathbb{R}^n)$.

3.5 More on Linear Hyperbolic Systems

Here, we consider additional properties of the linear wave equation not studied above, and also consider the more general class of *hyperbolic systems* [6]. As we shall see, this leads naturally to the notion of *propagation*. Some of these results are presented, perhaps with less exposition, in the Appendix.

⁵In the sense that for every $\phi, \psi \in C_c^\infty((-\infty, T) \times \mathbb{R}^n)$, $\langle L\phi, \psi \rangle = \langle \phi, L^* \psi \rangle$.

⁶ $\|uv\|_{L^1} \leq \|u\|_{W^{(-s,p)}} \|v\|_{W^{(-s,q)}}$, where p, q are conjugate exponents

⁷Complete proof that $(\phi, \partial_t \phi)$ belong to the appropriate spaces

Method of Characteristics We start with a simple setting: consider the homogeneous linear wave equation in $\mathbb{R}^{1+1}$, given by

$$\partial_t \phi + c(t, x) \partial_x \phi = 0,$$

where $c(t, x)$ is assumed to be a smooth real-valued function. We also consider the ordinary differential equation, $\dot{x}(t) = c(t, x(t))$. For a smooth function ϕ , one has that along the integral curves $(1, x(t))$,

$$\frac{d}{dt} \phi(t, x(t)) = (\partial_t \phi + c \partial_x \phi)|_{(t, x(t))}.$$

Thus it follows that the solutions of the homogeneous 1D linear wave equation are precisely those solutions ϕ that are constant along the integral curves $(t, x(t))$. We call these curves the **characteristic curves** of the partial differential operator $L = \partial_t + c(t, x) \partial_x$.

The **method of characteristics** then boils down to considering a differential equation in one spatial dimension, reducing the equation to an ODE on integral curves, and constructing an overall solution from the solutions on each integral curve (e.g., see Chapter 1 of [2] for an in-depth coverage of the subtleties and generalizations of the method of characteristics). It then follows that certain existence and uniqueness theorems follow easily from this method. E.g.,

Proposition 3.9 (Existence and Uniqueness for 1D Wave Equation). *If $c \in \mathbb{R}$ is constant, then $\phi \in C^\infty([0, T] \times \mathbb{R})$ satisfies*

$$\partial_t \phi + c \partial_x \phi = 0$$

if and only if there is an $f \in C^\infty(\mathbb{R})$ such that $\phi = f(x - ct)$. The function f is uniquely determined.

Proof. By the method of characteristics, for constant c , the characteristics along which u is constant are the lines $(t, x + ct)$. Thus, $u(t, x) = u(0, x - ct)$. If one sets $f(x) = u(0, x)$, then the forward direction is satisfied. Conversely, if there exists $f \in C^\infty$ such that $\phi(t, x) = f(x - ct)$, then

$$\partial_t \phi = -cf'(x - ct), \quad c \partial_x \phi = cf'(x - ct).$$

Thus, $\partial_t \phi + c \partial_x \phi = 0$. □

I.e., the Cauchy problem above is uniquely solvable by a C^∞ function, and the solutions are *propagating* waves translating rigidly with velocity c . We may also consider the same differential equation, but where $c(t, x)$ is *not* bounded.

Example 3.1 (Nonuniqueness for Unbounded Speed). *Consider the differential equation,*

$$\partial_t \phi + c(t, x) \partial_x \phi = 0,$$

where $c(t, x) = x^2$ is not bounded. Fix a point $(0, x_0)$ in spacetime. The characteristic through this point is the solution of the differential equation,

$$x' = x^2, \quad x(0) = x_0,$$

whence

$$x(t) = \frac{x_0}{1 - x_0 t}.$$

Inverting this, one obtains

$$x_0 = \frac{x}{1 + xt}.$$

Thus for each point (t, x) with $t \geq 0$, there is a unique characteristic tracing back to $t = 0$. Therefore, given initial data $\phi(0, x) = g(x)$, the solution $\phi(t, x)$ is uniquely determined in $t \geq 0$ by requiring ϕ to be constant on the characteristics. By plotting the integral curves, we observe the following: for $|x_0| \geq 1$, the corresponding integral curves all diverge before $t = 1$, for $x_0 = 1$, the integral curve passing through $(0, x_0)$ diverges at $t = 1$, and for $x_0 \in (0, 1)$, the integral curves blow up at finite time $t = 1/x_0 > 1$. The backward characteristic through $(0, -1)$ diverges to $-\infty$ at $t = -1$. These observations have the following consequences.

Since information about the data on $[1, \infty)$ blows up before $t = 1$, the data for $|x| \geq 1$ does not influence the solution in $t \geq 1$. Consequently, there has been a loss of information so that the initial values do not uniquely determine a solution in $t < 0$.

Thus, going forward, we shall make the following hypotheses:

Hypotheses 3.1 Suppose that for all $T > 0$, $\partial_{t,x}^\alpha c \in L^\infty([0, T] \times \mathbb{R})$, and that the coefficient $d(t, x)$ satisfies analogous bounds.

One can also generalize to *systems* of hyperbolic equations. For instance, the method of characteristics applies to systems of the form

$$u_t + \text{diag}(c_1(t, x), \dots, c_N(t, x))u = 0,$$

where $u = (u_1(t, x), \dots, u_N(t, x))^\top$, and each u_j is constant on characteristics with speed $c_j(t, x)$ that satisfy Hypotheses 3.1. More generally, the method of characteristics extends to systems given by the partial differential operator,

$$L = \partial_t + A(t, x)\partial_x + B(t, x),$$

where A, B are smooth matrix-valued functions such that for all $T \in \mathbb{R}$ and for all multiindices α , $\partial_{t,x}^\alpha A, \partial_{t,x}^\alpha B \in L^\infty([-T, T] \times \mathbb{R})$ when the following hypothesis is satisfied,

Hypothesis 3.2 There is smooth matrix-valued function $M(t, x)$ so that for all $T \in \mathbb{R}$, and all multiindices α ,

$$\partial_{t,x}^\alpha(M), \partial_{t,x}^\alpha(M^{-1}) \in L^\infty([0, T] \times \mathbb{R}),$$

and MAM^{-1} is diagonal and real.

Here, real diagonalizability of A just ensures that the linear operator remains hyperbolic-type, and the bounds on M and A just ensure that the characteristic speeds do not diverge (thus preventing the loss of information and/or uniqueness of solutions).

Definition 3.3 (Strictly Hyperbolic Systems). If for each (t, x) , the matrix A has N distinct real eigenvalues $c_1(t, x) < c_2(t, x) < \dots < c_N(t, x)$, the system is said to be **strictly hyperbolic**. To ensure that the estimates on M and M^{-1} are uniform as $|x| \rightarrow \infty$, we also make the additional assumption

$$\inf_{(t,x) \in [0,T] \times \mathbb{R}} \min_{2 \leq j \leq N} (c_j(t, x) - c_{j-1}(t, x)) > 0.$$

Explain?

A strictly hyperbolic system satisfies Hypothesis 3.2.

In particular, if the hypothesis holds, then since one has that $A = MDM^{-1}$ with $D = \text{diag}(c_1, \dots, c_N)$, we can write

$$L = \partial_t + MDM^{-1}\partial_x + B(x, t)$$

!! Complete Later !!

3.5.1 Energy Estimates

Here, we return again to energy estimates for *systems* of hyperbolic partial differential equations. As in the previous section, we will initially only consider first-order hyperbolic PDE.

Constant Coefficient Systems We start by considering the constant coefficient systems,

$$\partial_t \phi + \sum_{j=1}^d A_j \partial_{x^j} \phi = f,$$

where $\phi = (\phi_1, \dots, \phi_d)^\top$, and each A_j is a matrix-valued function. One has the following definition,

Definition 3.4 (Symmetric Hyperbolic Operator (Constant Coefficient)). *A constant coefficient operator*

$$L = \partial_t + \sum_{j=1}^d A_j \partial_{x^j}$$

on $\mathbb{R}^d$ is said to be **symmetric hyperbolic** if each A_j is hermitian symmetric.

Requiring that the A_j be hermitian *symmetric* is very useful for computing energy estimates. We recall that if A is symmetric, then the matrix-valued operator $\exp(-itA)$ is *unitary*. So now, computing the Fourier transform of the PDE $L\phi = 0$, one obtains,

$$\hat{\phi}(t, \xi) = \exp\left(-it \sum_{j=1}^d A_j \xi_j\right) \hat{\phi}(0, \xi).$$

But then, using the unitariness of the the exponential operator, we recover that for all $t \geq 0$,

$$|\hat{\phi}(t, \xi)|^2 = |\hat{\phi}(0, \xi)|^2.$$

In particular, integrating over $d\xi$ and using Plancherel's theorem, one easily obtains conservation of the L^2 -norm: $\|\phi(t)\|_{L^2(\mathbb{R}^d)} = \|\phi(0)\|_{L^2(\mathbb{R}^d)}$. Likewise, under appropriate assumptions on ϕ , one could also obtain conservation of the Sobolev norms.

As we saw before, these types of energy estimates segue easily into a discussion of (local) well-posedness, existence, and unique. Thus as an analogue to the existence and uniqueness results in the above subsections, we have the following result.

Proposition 3.10 (Existence and Uniqueness for Constant Coefficient Symmetric Hyperbolic PDE). *Suppose that $L = \partial_t + \sum_{j=1}^d A_j \partial_j$ is a constant coefficient symmetric hyperbolic operator. For $g \in H^\infty(\mathbb{R}^d)$, there exists a unique solution $\phi \in C^\infty(\mathbb{R}; H^\infty(\mathbb{R}^d))$ to the initial value problem*

$$L\phi = 0, \quad \phi|_{\{t=0\}} = g. \tag{13}$$

This solution is given by $\widehat{\phi}(t, \xi) = e^{-2\pi i t \sum A_j \xi_j} \hat{g}$. Furthermore, for all t, s , $\|\phi(t)\|_{H^s(\mathbb{R}^d)} = \|\phi(0)\|_{H^s(\mathbb{R}^d)}$.

Proof. We first prove existence of a solution: consider the function $\hat{\phi}(t, \xi) = \exp(-2\pi i t \sum A_j \xi_j) \hat{g}$. Then one notes that

$$|\hat{\phi}(t, \xi)|^2 = |\hat{g}|^2 \implies \|\hat{\phi}(t)\|_{L^2(\mathbb{R}^d)}^2 = \|\hat{g}\|_{L^2(\mathbb{R}^d)}^2 = \|g\|_{L^2(\mathbb{R}^d)}^2 < \infty,$$

where the last equality follows from Plancherel's theorem, and the final inequality follows from the fact that $g \in H^\infty(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$. Thus the inverse Fourier transform of $\hat{\phi}(t, \xi)$ is well-defined and also L^2 . In particular, one has

$$\phi(t, x) = \int_{\mathbb{R}^d} e^{2\pi i(x \cdot \xi - t \sum A_j \xi_j)} \hat{g}(\xi) d\xi.$$

It is immediately clear that $\phi(t, \cdot) \in C^\infty$. Now we must show that $\phi(\cdot, x) \in H^\infty(\mathbb{R}^d)$. Let $s \in \mathbb{R}$ be an arbitrary fixed real number. Then,

$$|(1 + |\xi|^2)^{s/2} \hat{\phi}(t, \xi)|^2 = (1 + |\xi|^2)^s \cdot |\hat{g}|^2.$$

⁸Note the 2π constant; [6] uses a slightly different convention for the Fourier transform.

But since we assumed that $g \in H^\infty(\mathbb{R}^d)$, it follows that $(1 + |\xi|^2)^{s/2} \hat{g}(\xi) \in L^2(\mathbb{R}^d)$. Thus, it follows that $\phi(\cdot, x) \in H^s(\mathbb{R}^d)$ for all $s \in \mathbb{R}$. Hence, $\phi \in C^\infty(\mathbb{R}; H^\infty(\mathbb{R}^d))$. To show that ϕ actually solves the PDE, it suffices to show that $\hat{\phi}$ solves the ODE obtained by taking the Fourier transform of the PDE. But we've already shown this to be the case. Therefore, the proof of existence is complete. Moreover, the above equality also establishes conservation of the Sobolev norms for all t, s . All that remains is to prove uniqueness. Assume that ϕ_1, ϕ_2 are two solutions to the initial value problem. Then their difference $\phi = \phi_1 - \phi_2$ also solves the PDE $L\phi = 0$, but with the initial value identically zero. Applying the above results (in particular, the Sobolev norm conservation), it follows that ϕ is identically zero. Thus the proof concludes. $\square$

Variable Coefficient Systems Now we shall consider variable coefficient systems, analogous to the variable coefficient equations studied in Section 3.4. We start with the following definition.

Definition 3.5 (Symmetric Hyperbolic Systems). *In $\mathbb{R}^{1+d}$ introduce the coordinates $y = (y_0, \dots, y_d)$, where $y_0 = t$, and for each $1 \leq j \leq d$, $y_j = x_j$. Then the partial differential operator,*

$$L(y, \partial) = \sum_{\mu=0}^d A_\mu(y) \frac{\partial}{\partial y_\mu} + B(y) \quad (14)$$

*is said to be **symmetric hyperbolic** if the following hold:*

1. *The coefficient matrices A_μ and B have uniformly bounded derivatives: for all α ,*

$$\sup_{y \in \mathbb{R}^{1+d}} \|\partial_y^\alpha (A_\mu(y), B(y))\| < \infty;$$

2. *A_μ are hermitian symmetric;*
3. *A_0 is strictly positive in the sense that there exists $c > 0$ so that for all y , $A_0(y) \geq cI$.*

It follows that this system satisfies an L^2 estimate of the following form.

Proposition 3.11. *L^2 Estimates For every $s \in \mathbb{R}$, there is a constant $C(s, L)$ so that for all $t \geq 0$ and all $u \in C^1(\mathbb{R}; H^s(\mathbb{R}^d)) \cap C(\mathbb{R}; H^{s+1}(\mathbb{R}^d))$,*

$$\|u(t)\|_{H^s(\mathbb{R}^d)} \leq C e^{Ct} \|u(0)\|_{H^s(\mathbb{R}^d)} + \int_0^t C e^{C(t-\sigma)} \|(Lu)(\sigma)\|_{H^s(\mathbb{R}^d)} d\sigma,$$

where L is a symmetric hyperbolic system.

Proof. Let us consider a few cases.

1. First assume that $A_0 = I$, and define

$$G(t) = \sum_j A_j(t, x) \partial_j + B(t, x),$$

and define its formal adjoint differential operator G^* by

$$G^*(t) = - \sum_j A_j^*(t, x) \partial_j + B(t, x)^* - \sum_j (\partial_j A_j^*)$$

so that for all $\phi, \psi \in C_c^\infty(\mathbb{R}^d)$, $\langle G(t)\phi, \psi \rangle = \langle \phi, G^*(t)\psi \rangle$. In particular, one has

$$G(t) + G^*(t) = B(t, x) + B^*(t, x) - \sum_{j=1}^d \partial_j A_j,$$

where we used the fact that since the A_j are hermitian symmetric, $A_j^* = A_j$. Therefore, $G(t) + G^*(t)$ is multiplication by a uniformly bounded matrix, which means that as an operator on $L^2(\mathbb{R}^d)$, $\|G(t) + G^*(t)\| \leq 2C$ for some $C > 0$. Note that this means that $G(t)$ is **nearly anti-self-adjoint**. If we assume that $u \in C(\mathbb{R}; H^1(\mathbb{R}^d))$, $u_t \in C(\mathbb{R}; L^2(\mathbb{R}^d))$, and $f := Lu$, then it is straightforward to see that $u'(t) + G(t)u(t) = f(t)$. $\square$

4 Local Theory for Nonlinear Wave Equations

4.1 Quasilinear Equations

In this subsection, we are most interested in the following quasilinear wave equations for $\phi : I \times \mathbb{R}^n \rightarrow \mathbb{R}$:

$$\begin{cases} \partial_\alpha (a^{\alpha\beta}(\phi) \partial_\beta \phi) = F(\phi, \partial\phi) \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n), \end{cases} \quad (15)$$

where $k \in \mathbb{N}$ is specified later, $a^{\alpha\beta}$ is symmetric and satisfies,

$$\sum_{\alpha, \beta} |a^{\alpha\beta} - m^{\alpha\beta}| \leq \frac{1}{10}, \quad a(0) = 0, \quad F(0, 0) = 0, \quad (*)$$

and $a^{\alpha\beta} : \mathbb{R} \rightarrow \mathbb{R}$ and $F : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$ are smooth functions. As we will see, unlike the linear wave equation, (15) is only *locally* (in time) well-posed. We will first state the main result of this section, then prove a couple of technical results that will then enable us to prove the theorem.

Theorem 4.1 (Local Well-Posedness of the Quasilinear Wave Equations). *Fix a and F satisfying assumptions $(*)$. Then the Cauchy problem (15) is locally well-posed in the following sense.*

1. (EXISTENCE AND UNIQUENESS OF LOCAL-IN-TIME SOLUTIONS) *There exists*

$$T = T(\|\phi_0\|_{H^{n+2}(\mathbb{R}^n)}, \|\phi_1\|_{H^{n+1}(\mathbb{R}^n)}) > 0$$

*such that there exists a **classical** solution ϕ to (15) with*

$$(\phi, \partial_t \phi) \in L^\infty([0, T]; H^{n+2}(\mathbb{R}^n)) \times L^\infty([0, T]; H^{n+1}(\mathbb{R}^n)).$$

Moreover, this solution is unique in the function space above.

2. (CONTINUOUS DEPENDENCE ON INITIAL DATA) *Let $\phi_0^{(i)}, \phi_1^{(i)}$ be sequences of functions such that $\phi_0^{(i)} \rightarrow \phi_0$ in $H^{n+2}(\mathbb{R}^n)$ and $\phi_1^{(i)} \rightarrow \phi_1$ in $H^{n+1}(\mathbb{R}^n)$ as $i \rightarrow \infty$. Then taking $T > 0$ sufficiently small, we have*

$$\|(\phi^{(i)} - \phi, \partial_t(\phi^{(i)} - \phi))\|_{L^\infty([0, T]; H^s(\mathbb{R}^n)) \times L^\infty([0, T]; H^{s-1}(\mathbb{R}^n))} \rightarrow 0$$

as $i \rightarrow \infty$ for every $1 \leq s < n + 2$.

A Explicit Representation Formulas

In this section, we work through the derivations of explicit representation formulas for initial value problems of the *wave equation*,

$$\square u := u_{tt} - \Delta u = 0 \quad (\text{or more generally, } f), \quad (16)$$

where the unknown is the function $u : \bar{U} \times [0, \infty) \rightarrow \mathbb{R}$, $u = u(x, t)$, with $x \in U^{\text{open}} \subseteq \mathbb{R}^n$ and $t > 0$, Δ is taken with respect to the spatial variables $x = (x_1, \dots, x_n)$, and $f : U \times [0, \infty) \rightarrow \mathbb{R}$ is a given function [1].

A.0.1 The Wave Equation in One Spatial Dimension – D'Alembert's Formula

The simplest example of the wave equation happens in the case $n = 1$. Consider the following initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & \text{in } \mathbb{R} \times (0, \infty) \\ u = g, u_t = h & \text{on } \mathbb{R} \times \{t = 0\}. \end{cases} \quad (17)$$

We note that the wave equation can be written in a more suggestive manner:

$$0 = u_{tt} - u_{xx} = (\partial_t + \partial_x)(\partial_t - \partial_x)u =: (\partial_t + \partial_x)v, \quad (18)$$

where we set $v = v(x, t) = u_t - u_x$. Thus, v satisfies the transport equation, $v_t + v_x = 0$. Using the solution to the transport equation in one dimension, one obtains the solution,

$$v(x, t) = a(x - t),$$

where $a(x) := v(x, 0)$. This gives us the nonhomogeneous transport equation,

$$u_t - u_x = a(x - t),$$

for which the explicit representation formula was shown (see Appendix) to be

$$u(x, t) = \int_0^t a(x + (t - s) - s) ds + g(x + t) = \frac{1}{2} \int_{x-t}^{x+t} a(y) dy + g(x + t). \quad (19)$$

It remains for us to identify the function $a(x, t)$. Using the definition of $v(x, t)$, at $t = 0$, one has,

$$a(x) = v(x, 0) = u_t(x, 0) - u_x(x, 0) = h(x) - g'(x).$$

Therefore, using the fundamental theorem of calculus and the above results, we conclude that

$$\boxed{u(x, t) = \frac{1}{2} [g(x + t) - g(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy} \quad (x \in \mathbb{R}^n, t \geq 0). \quad (20)$$

This explicit formula is called **d'Alembert's formula** for $n = 1$. Hence, we have shown that *if* the wave equation has a “sufficiently smooth solution u ”, then it must be given by d'Alembert's formula. The following theorem formalizes this result.

Theorem A.1 (Solution to the Wave Equation ($n = 1$)). *Assume $g \in C^2(\mathbb{R})$, $h \in C^1(\mathbb{R})$, and define u by d'Alembert's formula. Then*

- (i) $u \in C^2(\mathbb{R} \times [0, \infty))$.
- (ii) $u_{tt} - u_{xx} = 0$ in $\mathbb{R} \times (0, \infty)$
- (iii) $\lim_{\substack{(x,t) \rightarrow (x^0,0) \\ t>0}} u(x, t) = g(x^0)$ and $\lim_{\substack{(x,t) \rightarrow (x^0,0) \\ t>0}} u_t(x, t) = h(x^0)$ for each point $x^0 \in \mathbb{R}$.

Proof.

(i) Define u by d'Alembert's formula. Then we compute the following:

$$\begin{aligned} u_x &= \frac{1}{2}[g'(x+t) + g'(x-t)] + \frac{1}{2}[h(x+t) - h(x-t)], \\ u_t &= \frac{1}{2}[g'(x+t) - g'(x-t)] + \frac{1}{2}[h(x+t) + h(x-t)], \end{aligned}$$

whence

$$\begin{aligned} u_{xx} &= \frac{1}{2}[g''(x+t) + g''(x-t)] + \frac{1}{2}[h'(x+t) - h'(x-t)], \\ u_{xt} &= \frac{1}{2}[g''(x+t) - g''(x-t)] + \frac{1}{2}[h'(x+t) + h'(x-t)], \\ u_{tt} &= \frac{1}{2}[g''(x+t) + g''(x-t)] + \frac{1}{2}[h'(x+t) - h'(x-t)]. \end{aligned}$$

Since g', g'' and h, h' are all continuous at every $x \in \mathbb{R}$ and $t \in [0, \infty)$, as are the functions $x+t$ and $x-t$, we conclude that each of the first and second-order partial derivatives of u are continuous on $\mathbb{R} \times [0, \infty)$. Thus, $u \in C^2(\mathbb{R} \times [0, \infty))$.

(ii) From the above calculations, it follows immediately that $u_{tt} - u_{xx} = 0$.

(iii) By (1), u is continuous at every $(x, t) \in \mathbb{R} \times [0, \infty)$, and thus continuous at every point of the form $(x^0, 0) \in \mathbb{R} \times \{0\}$. Thus by the limit definition of continuity, one has

$$\lim_{(x,t) \rightarrow (x^0,0)} u(x,t) = u(x^0, 0) =: g(x^0).$$

Likewise, by our remark above, $u_t(x, t) \in C^1(\mathbb{R} \times [0, \infty))$. Thus by the limit definition of continuity, one has

$$\lim_{(x,t) \rightarrow (x^0,0)} u_t(x,t) = u_t(x^0, 0) =: h(x^0).$$

□

A.0.2 The Wave Equation in One Spatial Dimension – Half-Line

Now we will consider the modified initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R}_+ \times (0, \infty)) \\ u = g, u_t = h & (\text{in } \mathbb{R}_+ \times \{t = 0\}) \\ u = 0 & (\text{on } \{x = 0\} \times (0, \infty)). \end{cases} \quad (21)$$

To solve this problem, we will use the *method of reflections*; i.e., we will extend the initial value problem to the entire real line by means of the odd reflections:

$$\tilde{u}(x, t) = \begin{cases} u(x, t) & x \geq 0 \\ -u(-x, t) & x \leq 0. \end{cases}$$

$$\tilde{g}(x) = \begin{cases} g(x), & x \geq 0 \\ -g(-x), & x \leq 0, \end{cases}$$

$$\tilde{h}(x) = \begin{cases} h(x), & x \geq 0 \\ -h(-x), & x \leq 0. \end{cases}$$

It is straightforward to check that $\tilde{u}, \tilde{g}, \tilde{h}$ satisfy the ordinary wave equation ($n = 1$) on the whole real line. Thus by d'Alembert's formula, we obtain the following solution:

$$u(x, t) = \begin{cases} \frac{1}{2}[g(x+t) - g(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy & (x \geq t \geq 0) \\ \frac{1}{2}[g(x+t) - g(t-x)] + \frac{1}{2} \int_{t-x}^{x+t} h(y) dy & (0 \leq x \leq t) \end{cases} \quad (22)$$

A.0.3 The Wave Equation in $n \geq 2$ Spatial Dimension – Kirchoff's and Poisson's Formulas

In this subsection, we will develop explicit representation formulas for the wave equation for dimensions $n \geq 2$. To develop these formulas, we require the *method of spherical means*. Consider the following definitions.

Definition A.1 (Spherical Means).

(i) Let $x \in \mathbb{R}^n$, $t > 0$, and $r > 0$. We define the average of $u(\cdot, t)$ over the sphere $\partial B(x, r)$ to be

$$U(x; r, t) = \oint_{\partial B(x, r)} u(y, t) dS(y). \quad (23)$$

(ii) We make the analogous definitions,

$$\begin{cases} G(x; r) &= \oint_{\partial B(x, r)} g(y) dS(y). \\ H(x; r) &= \oint_{\partial B(x, r)} h(y) dS(y). \end{cases} \quad (24)$$

For any fixed x , we will regard U, G, H as functions of r and t .

Lemma A.1 (Euler-Poisson-Darboux Equation). Fix $x \in \mathbb{R}^n$, $n, m \geq 2$, and let $u \in C^m(\mathbb{R}^n \times [0, \infty))$ satisfy the initial value problem,

$$\begin{cases} u_{tt} - \Delta u = 0 & (\text{in } \mathbb{R}^n \times (0, \infty)) \\ u = g, u_t = h & (\text{in } \mathbb{R}^n \times \{t = 0\}). \end{cases}$$

Then $U \in C^m(\overline{\mathbb{R}_+} \times [0, \infty))$, and U_{tt} satisfies the Euler-Poisson-Darboux initial value problem,

$$\begin{cases} U_{tt} - U_{rr} - \frac{n-1}{r} U_r = 0 & (\text{in } \mathbb{R}_+ \times (0, \infty)) \\ U = G, U_t = H & (\text{on } \mathbb{R}_+ \times \{t = 0\}) \end{cases} \quad (25)$$

Proof. We begin by taking the partial derivative of the spherical mean U with respect to r . To do so, we rewrite

$$U(x; r, t) = \oint_{\partial B(x, r)} u(y, t) dS(y) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x, r)} u(y, t) dS(y) = \frac{1}{n\alpha(n)} \int_{\partial B(0, 1)} u(x + rz, t) dS(z).$$

Thus,

$$\begin{aligned} U_r(x; r, t) &= \frac{1}{n\alpha(n)} \frac{\partial}{\partial r} \int_{\partial B(0, 1)} u(x + rz, t) dS(z) \\ &= \frac{1}{n\alpha(n)} \int_{\partial B(0, 1)} Du(x + rz, t) \cdot z dS(z) \\ &= \frac{r^{1-n}}{n\alpha(n)} \int_{\partial B(x, r)} Du(y, t) \cdot \frac{y-x}{r} dS(y) \\ &= \frac{r}{n\alpha(n)r^n} \int_{\partial B(x, r)} \frac{\partial u}{\partial \nu} dS(y) \\ &= \frac{r}{n\alpha(n)r^n} \int_{B(x, r)} \Delta u dy = \frac{r}{n} \oint_{B(x, r)} \Delta u dy, \end{aligned}$$

where passing the derivative inside the integral in the second line is justified since u is sufficiently smooth (meaning that Du exists a.e. and is integrable on $\partial B(0, 1)$), and the final line follows from Green's formula. Now taking some large $R > 0$, and restricting to $0 < r \leq R/2$, one has that since $B(x, r) \subset \overline{B(x, R/2)}$ and Δu is continuous (so that the extreme value theorem applies),

$$\|\Delta u\|_{L^\infty(B(x, r))} \leq \|\Delta u\|_{L^\infty(B(x, R))} =: C < \infty.$$

Hence,

$$U_r(x; r, t) \leq \frac{C}{n} r \implies \lim_{r \rightarrow 0^+} U_r(x; r, t) = 0.$$

Now we will compute the second partial derivative of U with respect to r . We compute,

$$\begin{aligned} U_{rr}(x; r, t) &= \frac{1-n}{n\alpha(n)r^n} \int_{B(x, r)} \Delta u \, dy + \frac{r^{1-n}}{n\alpha(n)} \frac{\partial}{\partial r} \int_{B(x, r)} \Delta u(y, t) \, dy \\ &= \frac{1-n}{n} \int_{B(x, r)} \Delta u(y, t) \, dy + \frac{r^{1-n}}{n\alpha(n)} \int_{\partial B(x, r)} \Delta u(y, t) \, dS(y) \\ &= \int_{\partial B(x, r)} \Delta u(y, t) \, dS(y) + \left(\frac{1}{n} - 1\right) \int_{B(x, r)} \Delta u(y, t) \, dy, \end{aligned}$$

where the second line follows from the coarea formula (see Appendix). By applying similar reasoning to above, one computes that $\lim_{r \rightarrow 0^+} U_{rr}(x; r, t) = (1/n)\Delta u$. In this manner, repeatedly differentiating U , one sees that $U \in C^m(\mathbb{R}_+ \times [0, \infty))$. Now differentiating,

$$\begin{aligned} (n-1)r^{n-2}U_r + r^{n-1}U_{rr} &= \frac{d}{dr}(r^{n-1}U_r) \\ &= \frac{d}{dr} \frac{1}{n\alpha(n)} \int_{B(x, r)} u_{tt} \, dy \\ &= \frac{1}{n\alpha(n)} \int_{\partial B(x, r)} u_{tt} \, dS(y) \\ &= r^{n-1} \int_{\partial B(x, r)} u_{tt} \, dS(y) =: r^{n-1}U_{tt}, \end{aligned}$$

with the third line following from the coarea formula. □

Now we will derive the explicit representation formula for the 3-dimensional wave equation by reducing the system to the one-dimensional Euler-Poisson-Darboux equation and solving this system using the formula we derived for the one-dimensional half-line equation.

Kirchoff's Formula for the Wave Equation ($n = 3$) Suppose $u \in C^2(\mathbb{R}^3 \times [0, \infty))$ solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & (\text{in } \mathbb{R}^3 \times [0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R}^3 \times \{t = 0\}) \end{cases} \quad (26)$$

Define the functions, $\tilde{U} = rU(x; r, t)$, $\tilde{G}(r) = rG(x; r)$, and $\tilde{H}(r) = rH(x; r)$. Then we observe the following:

(i)

$$\begin{aligned} \tilde{U}_{tt} &= rU_{tt} = r \left[U_{rr} + \frac{2}{r} U_r \right] \\ &= rU_{rr} + 2U_r = (U + rU_r)_r = \tilde{U}_{rr}. \end{aligned}$$

(ii) On $\mathbb{R}_+ \times \{t = 0\}$, $U = G$ and $U_t = H$ so that $\tilde{U} = \tilde{G}$ and $\tilde{U}_t = \tilde{H}$.

(iii) For $r = 0$, $\partial B(x, 0) = \emptyset$ so that $\tilde{U} = 0$ on $\{r = 0\} \times (0, \infty)$.

In other words, we have shown that $\tilde{U}$ satisfies the following initial value problem,

$$\begin{cases} \tilde{U}_{tt} - \tilde{U}_{rr} = 0 & (\text{in } \mathbb{R}_+ \times (0, \infty)) \\ \tilde{U} = \tilde{G}, \tilde{U}_t = \tilde{H} & (\text{on } \mathbb{R}_+ \times \{t = 0\}) \\ \tilde{U} = 0 & (\text{on } \{r = 0\} \times (0, \infty)) \end{cases}$$

Using the solution for the one-dimensional wave equation on the half-line, for $0 \leq r \leq t$, one has

$$\tilde{U}(x; r, t) = \frac{1}{2}[\tilde{G}(r+t) - \tilde{G}(t-r)] + \frac{1}{2} \int_{t-r}^{r+t} \tilde{H}(y) dy.$$

On the other hand, one has that

$$\begin{aligned} u(x, t) &= \lim_{r \rightarrow 0^+} \frac{\tilde{U}(x; r, t)}{r} \\ &= \lim_{r \rightarrow 0^+} \left[\frac{\tilde{G}(r+t) - \tilde{G}(t-r)}{2r} + \frac{1}{2r} \int_{t-r}^{r+t} \tilde{H}(y) dy \right] \\ &= \tilde{G}'(t) + \tilde{H}(t). \end{aligned}$$

Recall again that

$$\begin{aligned} \tilde{G}'(t) &= \frac{\partial}{\partial t} \left(t \int_{\partial B(x, t)} g(y) dS(y) \right) \\ &= \frac{\partial}{\partial t} \left(\frac{1}{n\alpha(n)t} \int_{\partial B(x, t)} g(y) dS(y) \right) \\ &= -\frac{1}{n\alpha(n)t^2} \int_{\partial B(x, t)} g(y) dS(y) + \frac{1}{n\alpha(n)t} \frac{\partial}{\partial t} \left(t^2 \int_{\partial B(0, 1)} g(x+tz) dS(z) \right) \\ &= -\int_{\partial B(x, t)} g(y) dS(y) + \frac{2}{n\alpha(n)} \int_{\partial B(0, 1)} g(x+tz) dS(z) + \frac{1}{n\alpha(n)t} \int_{\partial B(x, t)} Dg(y) \cdot \left(\frac{y-x}{t} \right) dS(y) \\ &= \int_{\partial B(x, t)} g(y) dS(y) + \int_{\partial B(x, t)} Dg(y) \cdot (y-x) dS(y). \end{aligned} \tag{27}$$

Likewise, one has

$$\tilde{H}(t) = tH(t) = \int_{\partial B(x, t)} th(y) dS(y).$$

Therefore, we conclude that

$$\boxed{u(x, t) = \int_{\partial B(x, t)} th(y) + g(y) + Dg(y) \cdot (y-x) dS(y)} \tag{28}$$

This is **Kirchoff's formula** for the wave equation in dimension 3.

Poisson's Formula for the Wave Equation ($n = 2$) Assume $u \in C^2(\mathbb{R}^2 \times [0, \infty))$ solves the IVP for $n = 2$. Unlike for the $n = 3$ case, we will consider the 2-dimensional problem and “view” it as a 3-dimensional problem. Define $\bar{u}(x_1, x_2, x_3, t) = u(x_1, x_2, t)$ so that u satisfies the IVP

$$\begin{cases} \bar{u}_{tt} - \Delta \bar{u} = 0 & (\text{in } \mathbb{R}^3 \times (0, \infty)) \\ \bar{u} = \bar{g}, \bar{u}_t = \bar{h} & (\text{on } \mathbb{R}^3 \times \{t = 0\}), \end{cases}$$

where we define $\bar{g}$ and $\bar{h}$ analogously. For $x = (x_1, x_2) \in \mathbb{R}^2$, write $\bar{x} = (x_1, x_2, 0) \in \mathbb{R}^3$. Then by Kirchoff's formula, one has

$$u(x, t) = \bar{u}(\bar{x}, t) = \frac{\partial}{\partial t} \left(t \int_{\partial \bar{B}(\bar{x}, t)} \bar{g} d\bar{S} \right) + t \int_{\partial \bar{B}(\bar{x}, t)} \bar{h} d\bar{S},$$

where $\bar{B}(\bar{x}, t)$ denotes the ball in $\mathbb{R}^3$ with center $\bar{x}$, radius $t > 0$, and where $d\bar{S}$ denotes the 2-dimensional surface measure on $\partial\bar{B}(\bar{x}, t)$. But we can write

$$\int_{\partial\bar{B}(\bar{x}, t)} \bar{g} d\bar{S} = \frac{1}{4\pi t^2} \int_{\partial\bar{B}(\bar{x}, t)} \bar{g} d\bar{S} = \frac{2}{4\pi t^2} \int_{B(x, t)} g(y) (1 + |D\gamma(y)|^2)^{1/2} dy,$$

where $\gamma(y) = (t^2 - |y - x|^2)^{1/2}$ for $y \in B(x, t)$. The factor 2 comes from the two hemispheres of $\partial\bar{B}(\bar{x}, t)$. Since $(1 + |D\gamma|^2)^{1/2} = t(t^2 - |y - x|^2)^{-1/2}$, one has

$$\begin{aligned} \int_{\partial\bar{B}(\bar{x}, t)} \bar{g} d\bar{S} &= \frac{1}{2\pi t} \int_{B(x, t)} \frac{g(y)}{(t^2 - |y - x|^2)^{1/2}} dy \\ &= \frac{t}{2} \int_{B(x, t)} \frac{g(y)}{(t^2 - |y - x|^2)^{1/2}} dy. \end{aligned}$$

Using this and the coarea formula, one obtains

$$u(x, t) = \frac{1}{2} \frac{\partial}{\partial t} \left(t^2 \int_{B(x, t)} \frac{g(y)}{(t^2 - |y - x|^2)^{1/2}} dy \right) + \frac{t^2}{2} \int_{B(x, t)} \frac{h(y)}{(t^2 - |y - x|^2)^{1/2}} dy.$$

Using the same reasoning as above to pass the derivative inside the integral, one has

$$\frac{\partial}{\partial t} \left(t^2 \int_{B(x, t)} \frac{g(y)}{(t^2 - |y - x|^2)^{1/2}} dy \right) = t \int_{B(x, t)} \frac{g(y)}{(t^2 - |y - x|^2)^{1/2}} dy + t \int_{B(x, t)} \frac{Dg(y) \cdot (y - x)}{(t^2 - |y - x|^2)^{1/2}} dy.$$

Thus for $n = 2$, we obtain **Poisson's Formula**,

$$u(x, t) = \frac{1}{2} \int_{B(x, t)} \frac{tg(y) + t^2 h(y) + t Dg(y) \cdot (y - x)}{(t^2 - |y - x|^2)^{1/2}} dy \quad (x \in \mathbb{R}^2, t > 0) \quad (29)$$

Notice that in solving the wave equation for $n = 2$, we used the *method of descent*; i.e., we solved the problem for $n = 3$ first, and then dropped down to $n = 2$.

General Solutions for $n \geq 2$ Following similar extensive calculations as above (see Appendix for the work), we obtain the following solutions to the wave equation,

$$u(x, t) = \begin{cases} \frac{1}{\gamma(n)} \left[\left(\frac{\partial}{\partial t} \right) \left(\frac{1}{t} \frac{\partial}{\partial t} \right)^{\frac{n-3}{2}} \left(t^{n-2} \int_{\partial B(x, t)} g dS \right) + \left(\frac{1}{t} \frac{\partial}{\partial t} \right)^{\frac{n-3}{2}} \left(t^{n-2} \int_{\partial B(x, t)} h dS \right) \right] & \begin{matrix} n \geq 3 \\ n \text{ odd} \end{matrix} \\ \frac{1}{\beta(n)} \left[\left(\frac{\partial}{\partial t} \right) \left(\frac{1}{t} \frac{\partial}{\partial t} \right)^{\frac{n-2}{2}} \left(t^n \int_{B(x, t)} \frac{g(y)}{(t^2 - |y - x|^2)^{1/2}} dy \right) \right. \\ \left. + \left(\frac{1}{t} \frac{\partial}{\partial t} \right)^{\frac{n-2}{2}} \left(t^n \int_{B(x, t)} \frac{h(y)}{(t^2 - |y - x|^2)^{1/2}} dy \right) \right] & \begin{matrix} n \geq 2 \\ n \text{ even} \end{matrix} \end{cases} \quad (30)$$

where in the first equation $\gamma_n = 1 \cdot 3 \cdot \dots \cdot (n-2)$, and $\beta(n) = 2 \cdot 4 \cdot \dots \cdot (n-2) \cdot n$. The above solutions highlight an interesting contrast between odd-dimensional, and even-dimensional solutions. Note that if n is odd and $n \geq 3$, the initial data at a point $x \in \mathbb{R}^n$ only affect the solution u along the boundary $\{(y, t) \mid t > 0, |x - y| = t\}$ of the cone $C = \{(y, t) \mid t > 0, |x - y| < t\}$. On the other hand, if n is given, the initial data can affect the solution within all of C . Thus, a disturbance originating at x propagates along a sharp wavefront in odd dimensions, but continues to have effects even after the leading edge of the wavefront passes in even dimensions. This phenomenon is **Huygen's principle**.

A.1 Solution to the Nonhomogeneous Wave Equation

Now we will use the solutions from the previous sections to address the nonhomogeneous wave equation,

$$\begin{cases} u_{tt} - \Delta u = f & (\text{in } \mathbb{R}^n \times (0, \infty)) \\ u = 0, u_t = 0 & (\text{on } \mathbb{R}^n \times \{t = 0\}). \end{cases} \quad (31)$$

To solve this problem, we use Duhamel's principle⁹ by setting $u = u(x, t; s)$ to be the solution of

$$\begin{cases} u_{tt}(\cdot; s) - \Delta u(\cdot, s) = 0 & (\text{in } \mathbb{R}^n \times (0, \infty)) \\ u(\cdot; s) = 0, u_t(\cdot; s) = f(\cdot, s) & (\text{on } \mathbb{R}^n \times \{t = 0\}). \end{cases} \quad (32)$$

Thus, by Duhamel's principle,

$$u(x, t) := \int_0^t u(x, t; s) ds \quad (x \in \mathbb{R}^n, t \geq 0) \quad (33)$$

is a solution of the above nonhomogeneous wave equation. Consequently, the solution to the general inhomogeneous wave problem,

$$\begin{cases} u_{tt} - \Delta u = f & (\text{in } \mathbb{R}^n \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R}^n \times \{t = 0\}) \end{cases} \quad (34)$$

is equal to the sum of the solutions to the problems

$$\begin{cases} u_{tt} - \Delta u = 0 & (\text{in } \mathbb{R}^n \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R}^n \times \{t = 0\}) \end{cases} \quad \text{and} \quad \begin{cases} u_{tt} - \Delta u = f & (\text{in } \mathbb{R}^n \times (0, \infty)) \\ u = 0, u_t = 0 & (\text{on } \mathbb{R}^n \times \{t = 0\}) \end{cases} \quad (35)$$

A.1.1 Example 1: Solution to Nonhomogeneous Problem for $n = 1$

By d'Alembert's formula, one has

$$u(x, t; s) = \frac{1}{2} \int_{x-t+s}^{x+t+s} f(y, s) dy.$$

Thus,

$$u(x, t) = \frac{1}{2} \int_0^t \int_{x-t+s}^{x+t+s} f(y, s) dy ds = \frac{1}{2} \int_0^t \int_{x-s}^{x+s} f(y, t-s) dy ds \quad (x \in \mathbb{R}, t \geq 0).$$

A.1.2 Example 2: Solution to Nonhomogeneous Problem for $n = 3$

By Kirchoff's formula, one has

$$u(x, t; s) = (t-s) \oint_{\partial B(x, t-s)} f(y, s) dS.$$

This implies that

$$\begin{aligned} u(x, t) &= \int_0^t (t-s) \oint_{\partial B(x, t-s)} f(y, s) dS ds \\ &= \int_0^t (t-s) \frac{1}{4\pi(t-s)^2} \int_{\partial B(x, t-s)} f(y, s) dS ds \\ &= \frac{1}{4\pi} \int_0^t \int_{\partial B(x, t-s)} \frac{f(y, s)}{t-s} dS ds \\ &= \frac{1}{4\pi} \int_0^t \int_{\partial B(x, r)} \frac{f(y, t-r)}{r} dS dr. \end{aligned}$$

⁹See https://en.wikipedia.org/wiki/Duhamel%27s_principle for an explanation.

Hence, we conclude that

$$u(x, t) = \frac{1}{4\pi} \int_{B(x, t)} \frac{f(y, t - |y - x|)}{|y - x|} dy \quad (x \in \mathbb{R}^3, t \geq 0),$$

where the integrand on the right is called a *retarded potential*.

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